

IWoAT 2026

Lecture Notes — Days 1–5

An Introduction to Stable Homotopy Theory
and Brave New Algebra

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A few passages headed *Understanding / Intuition* are my own annotations. The notes have not been reviewed or approved by the lecturers.

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PART I

Day 1: Abstract Homotopy Theory

1 Weak Equivalences and Model Categories

1.1 Homotopical categories

Definition 1.1 (Homotopical category). A *homotopical category* is a category $\mathcal{C}$ together with a subcategory $W \subseteq \mathcal{C}$, whose morphisms are called *weak equivalences*, such that W contains every identity and satisfies the two-out-of-six property: whenever

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$$

are composable and both gf and hg lie in W , then f, g, h , and hgf all lie in W .

Two-out-of-six implies two-out-of-three, but not closure under retracts.

Definition 1.2 (Homotopical functor). Let $(\mathcal{C}, W_{\mathcal{C}})$ and $(\mathcal{D}, W_{\mathcal{D}})$ be homotopical categories. A functor $F: \mathcal{C} \rightarrow \mathcal{D}$ is *homotopical* if it sends every morphism of $W_{\mathcal{C}}$ to a morphism of $W_{\mathcal{D}}$.

A homotopical functor descends to the localization.

1.2 The model-category axioms

Definition 1.3 (Model category). A *model category* is a complete and cocomplete category $\mathcal{M}$ equipped with three classes of morphisms:

$$W \quad (\text{weak equivalences}), \quad C \quad (\text{cofibrations}), \quad F \quad (\text{fibrations}),$$

subject to the following axioms.

- (MC1) The class W satisfies two-out-of-three.
- (MC2) Each of W, C , and F is closed under retracts.
- (MC3) A cofibration has the left lifting property with respect to every acyclic fibration, and an acyclic cofibration has the left lifting property with respect to every fibration. Here *acyclic* means “also a weak equivalence.”
- (MC4) Every map factors as a cofibration followed by an acyclic fibration, and also as an acyclic cofibration followed by a fibration.

Equivalently, $(C, F \cap W)$ and $(C \cap W, F)$ are weak factorization systems. We also assume functorial factorizations when functorial replacements are needed.

The lifting axiom concerns every commutative square

$$\begin{array}{ccc} A & \longrightarrow & X \\ \downarrow i & \nearrow \ell & \downarrow p \\ B & \longrightarrow & Y. \end{array}$$

We write $i \square p$ when a diagonal lift ℓ exists for every such square. Thus the two required lifting relations are

$$C \square (F \cap W), \quad (C \cap W) \square F.$$

Lifts and factorizations need not be unique. Different replacements can be compared by weak equivalences.

Definition 1.4 (Fibrant and cofibrant objects). Let $\emptyset$ and $*$ denote the initial and terminal objects of $\mathcal{M}$. An object X is *cofibrant* if $\emptyset \rightarrow X$ is a cofibration, and it is *fibrant* if $X \rightarrow *$ is a fibration. It is *bifibrant* if it is both.

Example 1.5. In the standard Quillen model structure on topological spaces, weak equivalences are weak homotopy equivalences and fibrations are Serre fibrations. In the Kan–Quillen model structure on simplicial sets, weak equivalences are the maps whose geometric realizations are weak homotopy equivalences, cofibrations are monomorphisms, and fibrant objects are Kan complexes. These are the basic examples of the axioms [Hov99].

A map $p: E \rightarrow B$ of spaces is a *Serre fibration* if it has the homotopy lifting property for disks. Equivalently, for each $n \geq 0$ it has the right lifting property in

$$\begin{array}{ccc} D^n \times \{0\} & \xrightarrow{\quad} & E \\ \downarrow & \nearrow \text{dashed} & \downarrow p \\ D^n \times I & \xrightarrow{\quad} & B. \end{array}$$

Example 1.6 (The projective model structure on connective complexes). Let R be a ring and let $\mathbf{Ch}_{\geq 0}(R)$ denote nonnegatively graded chain complexes of left R -modules. Its projective model structure has quasi-isomorphisms as weak equivalences and maps that are surjective in every positive degree as fibrations. Cofibrations are exactly the maps with the left lifting property against acyclic fibrations. In particular, cofibrant complexes are built from projective modules.

Convention 1.7 (Arrow notation). In diagrams, $\rightarrowtail$ denotes a cofibration and $\twoheadrightarrow$ denotes a fibration. A superposed $\sim$ marks a weak equivalence. Thus an acyclic cofibration is drawn $\xrightarrow{\sim}\rightarrowtail$ and an acyclic fibration as $\xrightarrow{\sim}\twoheadrightarrow$. These are markings on the arrows, not extra maps.

1.3 Cofibrant and fibrant replacement

For replacements, factor the maps from the initial object and to the terminal object. First, factor $\emptyset \rightarrow X$ as a cofibration followed by an acyclic fibration:

$$\emptyset \rightarrowtail QX \xrightarrow[\sim]{q_X} X,$$

where QX is cofibrant. Factoring $X \rightarrow *$ as an acyclic cofibration followed by a fibration gives

$$X \xrightarrow[\sim]{r_X} RX \twoheadrightarrow *,$$

where RX is fibrant. With functorial choices, Q and R are the cofibrant and fibrant replacement functors.

Cofibrant replacement maps to X , while fibrant replacement maps from X .

Factoring $QX \rightarrow *$ as an acyclic cofibration $QX \rightarrow RQX$ followed by a fibration gives a bifibrant replacement. Thus every object is weakly equivalent to one in $\mathcal{M}^{\text{cf}}$; choices are unique only up to weak equivalence.

1.4 Cylinder objects, path objects, and homotopy

For an object X , factor the fold map as

$$X \amalg X \xrightarrow{(i_0, i_1)} \text{Cyl}(X) \xrightarrow{\sim} X.$$

A *left homotopy* from f to g , both maps $X \rightarrow Y$, is a map $H: \text{Cyl}(X) \rightarrow Y$ whose restrictions along i_0 and i_1 are f and g . Dually, a path object is a factorization

$$Y \xrightarrow{\sim} \text{Path}(Y) \xrightarrow{(p_0, p_1)} Y \times Y,$$

and a *right homotopy* $f \simeq g$ is a map $X \rightarrow \text{Path}(Y)$ with endpoints f and g .

For a cofibrant source and fibrant target, left and right homotopy agree and compute maps in the homotopy category. Need both assumptions here.

2 The Homotopy Category and the Calculus of Fractions

2.1 Localization

Definition 2.1 (Localization and homotopy category). Let $(\mathcal{C}, W)$ be a category with weak equivalences. Its *localization* is a functor

$$\gamma: \mathcal{C} \rightarrow \mathcal{C}[W^{-1}]$$

that sends every element of W to an isomorphism and is universal with this property. For every category $\mathcal{D}$, precomposition with γ is fully faithful. Its essential image consists of the functors $\mathcal{C} \rightarrow \mathcal{D}$ that send W to isomorphisms. For a model category $\mathcal{M}$, the localized category is denoted

$$\text{Ho}(\mathcal{M}) = \mathcal{M}[W^{-1}]$$

and called its *homotopy category*.

Give each weak equivalence $w: X \rightarrow Y$ a formal inverse. General maps are zigzags:

$$X = X_0 \longleftarrow X_1 \longrightarrow X_2 \longleftarrow \cdots \longrightarrow X_n = Y,$$

where every backward arrow is a weak equivalence.

When the class W satisfies an appropriate Ore or fraction condition, a zigzag may be shortened to a roof

$$\begin{array}{ccc} & X' & \\ w \swarrow & & \searrow f \\ X & & Y, \end{array} \quad \text{representing} \quad f w^{-1}.$$

Identify two roofs after a common refinement. This strict calculus does not work in every relative category. For model categories, there is a homotopical version.

Proposition 2.2 (Computing morphisms in $\text{Ho}(\mathcal{M})$). *Let $X, Y \in \mathcal{M}$. Choose a cofibrant replacement $QX \rightarrow X$ and a fibrant replacement $Y \rightarrow RY$. Then there is a natural bijection*

$$\text{Ho}(\mathcal{M})(X, Y) \cong \mathcal{M}(QX, RY) / \simeq,$$

where $\simeq$ is either left or right homotopy. In particular, the full subcategory $\mathcal{M}^{\text{cf}}$ of bifibrant objects, with homotopic maps identified, is equivalent to $\text{Ho}(\mathcal{M})$.

Proof sketch. The replacement maps become isomorphisms after localization, so every morphism $X \rightarrow Y$ is represented by a map $QX \rightarrow RY$. For a cofibrant source and a fibrant target, the model axioms identify zigzag equivalence with left or right homotopy. This gives the stated bijection [Hov99, Chapter 1]. $\square$

3 Derived Functors

3.1 Total derived functors

Definition 3.1 (Total derived functors). Assume $F: \mathcal{M} \rightarrow \mathcal{N}$ preserves weak equivalences between cofibrant objects. Its *total left derived functor* is computed by

$$\mathbf{L}F: \mathrm{Ho}(\mathcal{M}) \longrightarrow \mathrm{Ho}(\mathcal{N}), \quad \mathbf{L}F(X) = F(QX).$$

Assume instead that $G: \mathcal{M} \rightarrow \mathcal{N}$ preserves weak equivalences between fibrant objects. Its *total right derived functor* is computed by

$$\mathbf{R}G: \mathrm{Ho}(\mathcal{M}) \longrightarrow \mathrm{Ho}(\mathcal{N}), \quad \mathbf{R}G(X) = G(RX).$$

These formulas live in the homotopy category. Up to canonical natural isomorphism, they do not depend on the choices of Q and R .

There are comparison morphisms

$$(\mathbf{L}F)\gamma_{\mathcal{M}} \Longrightarrow \gamma_{\mathcal{N}}F, \quad \gamma_{\mathcal{N}}G \Longrightarrow (\mathbf{R}G)\gamma_{\mathcal{M}},$$

coming from $QX \rightarrow X$ and $X \rightarrow RX$ when the point-set comparisons are defined.

3.2 Quillen adjunctions

Definition 3.2 (Quillen adjunction). An adjunction

$$\mathcal{M} \begin{array}{c} \xrightarrow{F} \\ \xleftarrow{G} \end{array} \mathcal{N} \quad (F \dashv G)$$

is a *Quillen adjunction* if F preserves cofibrations and acyclic cofibrations. Equivalently, G preserves fibrations and acyclic fibrations. It is a *Quillen equivalence* if, for every cofibrant $X \in \mathcal{M}$ and fibrant $Y \in \mathcal{N}$, a map $FX \rightarrow Y$ is a weak equivalence exactly when its adjoint $X \rightarrow GY$ is a weak equivalence.

Proposition 3.3 (Derived adjunction). *A Quillen adjunction $F \dashv G$ induces an adjunction*

$$\mathrm{Ho}(\mathcal{M}) \begin{array}{c} \xrightarrow{\mathbf{L}F} \\ \xleftarrow{\mathbf{R}G} \end{array} \mathrm{Ho}(\mathcal{N}).$$

If the original adjunction is a Quillen equivalence, this derived adjunction is an equivalence of categories.

4 Diagram Categories and Homotopy (Co)limits

4.1 Model structures on diagram categories

For a small category I , write $\mathcal{M}^I = \text{Fun}(I, \mathcal{M})$. Weak equivalences of diagrams are defined objectwise.

Definition 4.1 (Projective and injective diagram structures). When they exist, the *projective model structure* on $\mathcal{M}^I$ has objectwise weak equivalences and objectwise fibrations; its cofibrations are then determined by the left lifting property. The *injective model structure* has objectwise weak equivalences and objectwise cofibrations; its fibrations are determined by the right lifting property.

For small I and combinatorial $\mathcal{M}$, both structures exist. Reedy structures are another option. Cofibrancy and fibrancy apply to the whole diagram, including the structure maps.

Understanding / Intuition

Cofibrant entries alone do not ensure a projectively cofibrant diagram. Need to replace the structure maps too.

The ordinary adjunctions are

$$\mathcal{M}^I \begin{array}{c} \xrightarrow{\text{colim}_I} \\ \xleftarrow{\Delta} \quad \xrightarrow{\quad} \\ \xrightarrow{\quad} \\ \xrightarrow{\text{lim}_I} \end{array} \mathcal{M} \quad \text{colim}_I \dashv \Delta \dashv \text{lim}_I, \quad (4.1)$$

where Δ sends an object to the constant I -diagram. For the projective model structure, Δ preserves fibrations and acyclic fibrations, so $\text{colim}_I \dashv \Delta$ is a Quillen adjunction. For the injective model structure, Δ preserves cofibrations and acyclic cofibrations, so $\Delta \dashv \text{lim}_I$ is a Quillen adjunction.

Definition 4.2 (Homotopy colimit and homotopy limit). The *homotopy colimit* and *homotopy limit* are the total derived functors

$$\text{hocolim}_I D := \mathbf{L} \text{colim}_I(D), \quad \text{holim}_I D := \mathbf{R} \text{lim}_I(D).$$

If $Q_I D \rightarrow D$ is a projectively cofibrant replacement and $D \rightarrow R_I D$ is an injectively fibrant replacement, these are computed by

$$\text{hocolim}_I D \simeq \text{colim}_I Q_I D, \quad \text{holim}_I D \simeq \text{lim}_I R_I D. \quad (4.2)$$

The right-hand sides are point-set models for these homotopy (co)limits.

An ordinary colimit is characterized by

$$\mathcal{M}(\text{colim}_I D, Z) \cong \mathcal{M}^I(D, \Delta Z).$$

and the limit is defined dually.

4.2 Pushouts as homotopy colimits

Let I be the span category with three objects and two nonidentity arrows $0 \rightarrow 1$ and $0 \rightarrow 2$. An I -diagram is a span

$$\begin{array}{ccc} & A & \\ f \swarrow & & \searrow g \\ B & & C. \end{array}$$

Its ordinary colimit is the pushout $B \amalg_A C$:

$$\begin{array}{ccc} A & \xrightarrow{g} & C \\ f \downarrow & & \downarrow \\ B & \longrightarrow & B \amalg_A C. \end{array}$$

Thus the homotopy pushout is the homotopy colimit of this span:

$$B \amalg_A^h C := \operatorname{hocolim}(B \leftarrow A \rightarrow C).$$

For a cofibrant span, the ordinary pushout computes the homotopy pushout.

For spaces, use the double mapping cylinder:

$$B \amalg_{A \times \{0\}} (A \times I) \amalg_{A \times \{1\}} C. \quad (4.3)$$

Collapsing the interval gives the ordinary pushout, but need not be a weak equivalence without cofibrancy.

Example 4.3. For the span $* \leftarrow A \rightarrow *$, the ordinary pushout is a point, whereas the homotopy pushout is the unreduced suspension ΣA .

4.3 Pullbacks as homotopy limits

Dually, the limit of a cospan $B \xrightarrow{f} D \xleftarrow{g} C$ is the pullback

$$\begin{array}{ccc} B \times_D C & \longrightarrow & C \\ \downarrow & & \downarrow g \\ B & \xrightarrow{f} & D. \end{array}$$

For sets,

$$B \times_D C = \{(b, c) \mid f(b) = g(c)\}.$$

The homotopy pullback is the derived limit

$$B \times_D^h C := \operatorname{holim}(B \rightarrow D \leftarrow C).$$

For spaces, after the usual fibrant replacements, a model is

$$\{(b, \gamma, c) \mid \gamma: I \rightarrow D, \gamma(0) = f(b), \gamma(1) = g(c)\}. \quad (4.4)$$

For a fibrant cospan, the ordinary pullback computes the homotopy pullback. In a right-proper model category, the usual one-leg fibration formula also applies after fibrant replacement.

5 Simplicial Sets and Simplicial Enrichment

5.1 Simplicial sets

Let Δ be the simplex category. Its objects are the finite nonempty ordered sets

$$[n] = \{0 < 1 < \cdots < n\}, \quad n \geq 0,$$

and its morphisms are order-preserving maps.

Definition 5.1 (Simplicial set). A *simplicial set* is a functor

$$X: \Delta^{\text{op}} \longrightarrow \mathbf{Set}.$$

The set $X_n = X([n])$ is the set of n -simplices of X . The elementary injective and surjective maps in Δ induce face maps $d_i: X_n \rightarrow X_{n-1}$ and degeneracy maps $s_i: X_n \rightarrow X_{n+1}$, satisfying the simplicial identities.

The standard n -simplex is the representable simplicial set

$$\Delta[n]_m = \text{Hom}_{\Delta}([m], [n]).$$

By Yoneda, maps $\Delta[n] \rightarrow X$ are the elements of X_n . The boundary $\partial\Delta[n]$ is the union of all proper faces. For $0 \leq k \leq n$, the k th horn $\Lambda^k[n]$ is the union of all codimension-one faces except the k th:

$$\Lambda^k[n] \subseteq \partial\Delta[n] \subseteq \Delta[n].$$

A map $\Lambda^k[n] \rightarrow X$ specifies all but one face of a possible n -simplex. In the Kan–Quillen model structure, cofibrations are monomorphisms and fibrant objects are Kan complexes.

5.2 Categories enriched in simplicial sets

Definition 5.2 (Simplicially enriched category). A category $\mathcal{C}$ enriched in simplicial sets has a class of objects and, for each $X, Y \in \mathcal{C}$, a simplicial set $\text{Map}_{\mathcal{C}}(X, Y)$, together with composition and unit maps

$$\text{Map}_{\mathcal{C}}(Y, Z) \times \text{Map}_{\mathcal{C}}(X, Y) \longrightarrow \text{Map}_{\mathcal{C}}(X, Z), \quad \Delta[0] \longrightarrow \text{Map}_{\mathcal{C}}(X, X),$$

satisfying associative and unital laws as maps of simplicial sets.

The underlying ordinary category has

$$\text{Hom}_{\mathcal{C}}(X, Y) = \text{Map}_{\mathcal{C}}(X, Y)_0.$$

A 1-simplex relates two maps $X \rightarrow Y$. Higher simplices give coherent relations between these. Still need fibrancy conditions for homotopy theory.

5.3 Tensor, cotensor, and the SM7 axiom

Definition 5.3 (Tensor and cotensor). A simplicially enriched category $\mathcal{M}$ is *tensor*ed and *cotensor*ed over $\mathbf{sSet}$ if there are objects $X \otimes K$ and Y^K for $X, Y \in \mathcal{M}$ and $K \in \mathbf{sSet}$, natural in all variables, with natural bijections

$$\mathcal{M}(X \otimes K, Y) \cong \mathbf{sSet}(K, \text{Map}_{\mathcal{M}}(X, Y)) \cong \mathcal{M}(X, Y^K). \quad (5.1)$$

For cofibrant X , $X \otimes \Delta[1]$ is a cylinder object; for fibrant Y , $Y^{\Delta[1]}$ is a path object. A 1-simplex of $\text{Map}_{\mathcal{M}}(X, Y)$ is a map $X \otimes \Delta[1] \rightarrow Y$.

Definition 5.4 (Simplicial model category and SM7). A *simplicial model category* is a model category $\mathcal{M}$ that is enriched, tensored, and cotensored over $\mathbf{sSet}$, such that the enrichment is compatible with the model structure as follows. Given a cofibration $i: A \rightarrow B$ and a fibration $p: X \rightarrow Y$, the induced map

$$\text{Map}_{\mathcal{M}}(B, X) \longrightarrow \text{Map}_{\mathcal{M}}(A, X) \times_{\text{Map}_{\mathcal{M}}(A, Y)} \text{Map}_{\mathcal{M}}(B, Y) \quad (5.2)$$

is a fibration of simplicial sets. It is an acyclic fibration if either i or p is also a weak equivalence. This condition is traditionally called *SM7*.

The fiber product in (5.2) says that this square commutes:

$$\begin{array}{ccc} A & \longrightarrow & X \\ i \downarrow & & \downarrow p \\ B & \longrightarrow & Y. \end{array}$$

Equivalently, for a cofibration $i: A \rightarrow B$ in $\mathcal{M}$ and a cofibration $j: K \rightarrow L$ in $\mathbf{sSet}$, the pushout-product map

$$(B \otimes K) \amalg_{A \otimes K} (A \otimes L) \longrightarrow B \otimes L \quad (5.3)$$

is a cofibration in $\mathcal{M}$, acyclic if either i or j is acyclic. The two formulations correspond under (5.1).

Proposition 5.5 (Mapping spaces between good objects). *If A is cofibrant and X is fibrant in a simplicial model category $\mathcal{M}$, then $\text{Map}_{\mathcal{M}}(A, X)$ is a Kan complex. Moreover,*

$$\text{Ho}(\mathcal{M})(A, X) \cong \pi_0 \text{Map}_{\mathcal{M}}(A, X).$$

For arbitrary A and X , a derived mapping space is

$$\text{Map}_{\mathcal{M}}^h(A, X) := \text{Map}_{\mathcal{M}}(QA, RX),$$

well defined up to weak equivalence of simplicial sets.

Proof sketch. Apply SM7 to $\emptyset \rightarrow A$ and $X \rightarrow *$. This makes $\text{Map}(A, X) \rightarrow *$ a Kan fibration. Taking components and using Proposition 2.2 gives the bijection. $\square$

5.4 Simplicial localization

The Dwyer–Kan simplicial localization builds mapping spaces from weak equivalences.

Construction 5.6 (Hammock localization). For a category $(\mathcal{C}, W)$ with weak equivalences, the hammock localization $L_W^H \mathcal{C}$ is a simplicially enriched category with the same objects as $\mathcal{C}$. Its mapping simplicial sets are built from grids of zigzags in which the arrows directed backwards and all vertical arrows lie in W . Composition concatenates the grids and then applies the relations and refinements of the hammock construction.

Taking connected components gives ordinary localization:

$$\pi_0 \text{Map}_{L_W^H \mathcal{C}}(X, Y) \cong \mathcal{C}[W^{-1}](X, Y).$$

For a simplicial model category, the hammock mapping space is weakly equivalent to the derived mapping space:

$$\text{Map}_{L_W^H \mathcal{M}}(X, Y) \simeq \text{Map}_{\mathcal{M}}(QX, RY).$$

Simplicial localization also retains the derived mapping spaces.

6 $(\infty, 1)$ -Categories and Quasicategories

6.1 Models of $(\infty, 1)$ -categories

Definition 6.1 ($(\infty, 1)$ -category). An $(\infty, 1)$ -category is a higher category with objects, 1-morphisms, 2-morphisms, and so on, in which every k -morphism for $k > 1$ is invertible up to still higher morphisms. The 1-morphisms need not be invertible.

Complete Segal spaces, simplicial categories, and quasicategories are standard models.

An ordinary category has discrete mapping spaces. A Kan complex models an $(\infty, 1)$ -category in which every 1-morphism is invertible, so it is an ∞ -groupoid.

There are fully faithful inclusions

$$\mathbf{Cat} \xrightarrow{N} \mathbf{qCat} \xleftarrow[\text{all edges invertible}]{\text{all edges invertible}} \mathbf{Kan}.$$

6.2 Inner horns and coherent composition

Definition 6.2 (Quasicategory). A *quasicategory* is a simplicial set $\mathcal{C}$ such that every inner horn has a filler. Explicitly, for every $n \geq 2$ and $0 < k < n$, every solid diagram

$$\begin{array}{ccc} \Lambda^k[n] & \longrightarrow & \mathcal{C} \\ \downarrow & \nearrow \text{dashed} & \\ \Delta[n] & & \end{array}$$

admits a dashed lift. No uniqueness of the filler is required.

Vertices are objects, edges are morphisms, and a K -shaped diagram is a map $K \rightarrow \mathcal{C}$. The functor quasicategory is

$$\mathrm{Fun}(K, \mathcal{C}) = \mathcal{C}^K, \quad (\mathcal{C}^K)_n = \mathbf{sSet}(K \times \Delta[n], \mathcal{C}).$$

When $\mathcal{C}$ is a quasicategory, so is $\mathcal{C}^K$.

First case: $\Lambda^1[2]$. A map from this inner horn gives two composable edges:

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ & \searrow h & \downarrow g \\ & & z. \end{array}$$

A filler gives a composite $h: x \rightarrow z$ and a 2-simplex with boundary (f, g, h) . Higher inner horns give the coherence data.

Inner horns describe composition; outer horns describe inverse-like data.

Proposition 6.3. *Every Kan complex is a quasicategory. A quasicategory is a Kan complex exactly when all of its 1-morphisms are equivalences. If $\mathcal{A}$ is an ordinary category, its nerve $N\mathcal{A}$ is a quasicategory with unique inner-horn fillers; it is a Kan complex exactly when $\mathcal{A}$ is a groupoid.*

6.3 Mapping spaces, the core, and the homotopy category

Write $\mathrm{Map}_{\mathcal{C}}(x, y)$ for any of the equivalent Kan-complex models of the mapping space.

Definition 6.4 (Homotopy category of a quasicategory). For a quasicategory $\mathcal{C}$, its homotopy category $h\mathcal{C}$ has the vertices of $\mathcal{C}$ as objects and

$$\mathrm{Hom}_{h\mathcal{C}}(x, y) = \pi_0 \mathrm{Map}_{\mathcal{C}}(x, y)$$

as morphism sets. Composition is induced by filling inner 2-horns. The higher horn conditions make the resulting composition associative and unital on homotopy classes.

An edge is an *equivalence* if its image in $h\mathcal{C}$ is an isomorphism.

Definition 6.5 (Core). The *core* $\mathcal{C}^{\simeq}$ of a quasicategory $\mathcal{C}$ is the maximal Kan subcomplex containing every object and precisely the edges that are equivalences. It is the underlying ∞ -groupoid, or homotopy type, of objects and equivalences in $\mathcal{C}$.

For an ordinary category, the core is the nerve of its maximal subgroupoid; for a Kan complex X , it is all of X .

6.4 The Joyal model structure

Theorem 6.6 (Joyal model structure). *There is a model structure on $\mathbf{sSet}$ in which the cofibrations are the monomorphisms, the fibrant objects are the quasicategories, and the weak equivalences are the categorical equivalences. Between quasicategories this means essential surjectivity on homotopy categories together with weak equivalences on every mapping space; the Joyal notion extends this class to arbitrary simplicial sets. It is called the Joyal model structure.*

6.5 The underlying ∞ -category of a model category

Given a model category $(\mathcal{M}, W)$, take its simplicial localization and then a quasicategorical model. If $\mathcal{M}$ is already a simplicial model category, the full simplicial subcategory $\mathcal{M}^{\mathrm{cf}}$ has Kan mapping spaces by Proposition 5.5. Its homotopy-coherent nerve is a quasicategory. Write the resulting localization as $\mathcal{M}_{\infty}$.

Proposition 6.7 (Comparison with the model category). *For a simplicial model category $\mathcal{M}$ and objects $X, Y \in \mathcal{M}$ viewed in the localization, there are natural equivalences of homotopy types*

$$\mathrm{Map}_{\mathcal{M}_{\infty}}(X, Y) \simeq \mathrm{Map}_{\mathcal{M}}(QX, RY),$$

and an equivalence of ordinary categories

$$h(\mathcal{M}_{\infty}) \simeq \mathrm{Ho}(\mathcal{M}).$$

Even without a chosen simplicial enrichment, simplicial localization gives the model category an underlying ∞ -category.

6.6 Joins, cones, and ∞ -categorical limits

Joins give the cone shapes

$$K^{\triangleleft} = \Delta[0] \star K, \quad K^{\triangleright} = K \star \Delta[0].$$

Let $p: K \rightarrow \mathcal{C}$ be a diagram. A cone on p with vertex x is an extension

$$\begin{array}{ccc} K & \xrightarrow{p} & \mathcal{C} \\ \downarrow & \nearrow \bar{p} & \\ K^{\triangleleft} & & \end{array}$$

whose new vertex is sent to x . A cocone is defined similarly using $K^{\triangleright}$. The quasicategory of cones is the fiber

$$\text{Cone}(p) = \text{Fun}(K^{\triangleleft}, \mathcal{C}) \times_{\text{Fun}(K, \mathcal{C})} \{p\},$$

and similarly for cocones.

Definition 6.8 (∞ -categorical limit and colimit). A *limit cone* for p is a terminal object of $\text{Cone}(p)$. Its vertex is denoted $\lim_K p$. A *colimit cocone* is an initial object of the quasicategory of cocones, and its vertex is denoted $\text{colim}_K p$.

Equivalently, natural equivalences of mapping spaces characterize a limit object ℓ and a colimit object c :

$$\text{Map}_{\mathcal{C}}(x, \ell) \simeq \text{Map}_{\text{Fun}(K, \mathcal{C})}(\Delta x, p), \quad (6.1)$$

$$\text{Map}_{\mathcal{C}}(c, y) \simeq \text{Map}_{\text{Fun}(K, \mathcal{C})}(p, \Delta y). \quad (6.2)$$

Here Δx and Δy are constant diagrams. Taking π_0 gives the universal property for homotopy-coherent cones. This need not be an ordinary limit in $h\mathcal{C}$.

For the span shape, an ∞ -categorical colimit is a homotopy-coherent pushout. For the cospan shape, an ∞ -categorical limit is a homotopy-coherent pullback. If $\mathcal{M}$ presents $\mathcal{M}_{\infty}$ and the relevant diagram model structures exist, these constructions agree:

$$\text{colim}_{\mathcal{M}_{\infty}} D \simeq \text{hocolim}_{\mathcal{M}} D, \quad \lim_{\mathcal{M}_{\infty}} D \simeq \text{holim}_{\mathcal{M}} D.$$

Compare (4.2).

PART II

Day 2: Spectra and Stable Categories

7 Sequential Spectra and Smash Products

7.1 Sequential spectra

For pointed spaces there is an adjunction

$$\Sigma = - \wedge S^1 : \mathbf{Top}_* \rightleftarrows \mathbf{Top}_* : \Omega = \mathrm{Map}_*(S^1, -). \quad (7.1)$$

Definition 7.1 (Sequential spectrum). A *sequential spectrum* X of pointed spaces consists of pointed spaces X_n , for $n \geq 0$, and structure maps

$$\sigma_n : S^1 \wedge X_n \longrightarrow X_{n+1}.$$

A morphism $f : X \rightarrow Y$ is a sequence of pointed maps $f_n : X_n \rightarrow Y_n$ commuting with the structure maps. The adjoints

$$\tilde{\sigma}_n : X_n \longrightarrow \Omega X_{n+1}$$

give the same data.

For a pointed space K , the suspension spectrum is

$$(\Sigma^\infty K)_n = S^n \wedge K,$$

with the evident structure maps. The *sphere spectrum*

$$\mathbb{S} = \Sigma^\infty S^0 = (S^0, S^1, S^2, \dots)$$

is the stable unit. An Ω -spectrum has every adjoint structure map $X_n \rightarrow \Omega X_{n+1}$ a weak equivalence.

The stable homotopy groups of a sequential spectrum are defined by

$$\pi_k X = \mathrm{colim}_n \pi_{k+n}(X_n), \quad (7.2)$$

using the structure maps for the transition maps. Stable equivalences induce isomorphisms on these groups. For an Ω -spectrum the colimit stabilizes.

If E is a suitably fibrant Ω -spectrum, define, on finite pointed CW complexes,

$$E^q(K) = [\Sigma^\infty K, \Sigma^q E]_{\mathrm{st}}.$$

For $q \geq 0$ this is $[K, E_q]_*$. Conversely, use Brown representability and the suspension isomorphisms to get an Ω -spectrum from a reduced cohomology theory.

7.2 Pointed smash products and reduced suspension

For pointed spaces (X, x_0) and (Y, y_0) ,

$$X \wedge Y = \frac{X \times Y}{X \times \{y_0\} \cup \{x_0\} \times Y}. \quad (7.3)$$

Taking $Y = S^1 = I/\partial I$ gives the reduced suspension

$$\Sigma X = X \wedge S^1 \cong \frac{X \times I}{X \times \{0, 1\} \cup \{x_0\} \times I}. \quad (7.4)$$

In particular, $S^m \wedge S^n \cong S^{m+n}$. Use this to iterate the structure maps.

7.3 The smash-product problem

A ring spectrum should have a multiplication

$$\mu: R \wedge R \longrightarrow R$$

and a unit $\mathbb{S} \rightarrow R$.

First try: levelwise smash,

$$(X \wedge_{\text{lev}} Y)_n = X_n \wedge Y_n,$$

has the wrong grading. Graded convolution fixes this. Still need to handle permutations of suspension coordinates. For diagram spectra, these go into the indexing category [MMSS01].

8 Diagram Spectra

8.1 Enriched indexing categories

Let $\mathcal{V}$ be pointed spaces or pointed simplicial sets. A diagram spectrum is an enriched functor

$$X: \mathcal{D} \longrightarrow \mathcal{V}$$

for a suitable enriched indexing category $\mathcal{D}$.

For sequential spectra, use the linear indexing category $\mathcal{H}$. Its objects are the nonnegative integers. Schematically,

$$\mathcal{H}(m, n) = \begin{cases} S^{n-m}, & m \leq n, \\ *, & m > n. \end{cases} \quad (8.1)$$

Composition is induced by $S^{n-m} \wedge S^{m-\ell} \cong S^{n-\ell}$. An enriched functor $X: \mathcal{H} \rightarrow \mathbf{Top}_*$ gives evaluation maps

$$\mathcal{H}(m, n) \wedge X_m \longrightarrow X_n.$$

For $n = m + 1$, these are the structure maps. For larger gaps, they are coherent iterates of those maps.

Here m, n are objects of $\mathcal{H}$. The object $\mathcal{H}(m, n)$ is a based mapping space, not an ordinary hom-set.

Suppose $\mathcal{D}$ is a small symmetric monoidal $\mathcal{V}$ -category. Day convolution is defined by the coend

$$(X \star Y)(d) = \int^{a,b \in \mathcal{D}} \mathcal{D}(a \otimes b, d) \otimes X(a) \otimes Y(b). \quad (8.2)$$

Its unit is $\mathcal{D}(\mathbb{1}, -)$, and the internal hom is characterized by

$$[\mathcal{D}, \mathcal{V}](X \star Y, Z) \cong [\mathcal{D}, \mathcal{V}](X, \underline{\text{Hom}}(Y, Z)).$$

8.2 Coends, tensors, and Kan extension

A coend is a colimit with relations from both variables. For an ordinary small category $\mathcal{C}$ and $F: \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathcal{V}$, it can be computed as

$$\int^{c \in \mathcal{C}} F(c, c) = \text{coeq} \left(\prod_{u: c \rightarrow c'} F(c', c) \rightrightarrows \prod_c F(c, c) \right). \quad (8.3)$$

The two arrows use u in the two variables.

If a ring R is regarded as a one-object category enriched in abelian groups, a right R -module M and a left R -module N form the coend

$$M \otimes_R N \cong \int^* M(*) \otimes_{\mathbb{Z}} N(*).$$

The relation $(mr, n) \sim (m, rn)$ is the balancing relation.

Use the same coend for enriched left Kan extension. If $i: \mathcal{D} \rightarrow \mathcal{E}$ is enriched, then

$$(\text{Lan}_i X)(e) = \int^{d \in \mathcal{D}} \mathcal{E}(id, e) \otimes X(d). \quad (8.4)$$

It is left adjoint to restriction i^* whenever the coend exists.

For representables, apply enriched Yoneda: the coend in (8.4) reduces to evaluation.

8.3 Symmetric spectra

Definition 8.1 (Symmetric spectrum). A *symmetric spectrum* X in pointed spaces consists of a pointed Σ_n -space X_n for every $n \geq 0$ and equivariant structure maps

$$\sigma_n: X_n \wedge S^1 \longrightarrow X_{n+1},$$

such that the iterated map

$$X_n \wedge S^p \longrightarrow X_{n+p} \quad (8.5)$$

is $\Sigma_n \times \Sigma_p$ -equivariant, where the product subgroup sits in Σ_{n+p} by block permutations.

The Σ_n -action tracks permutations of the suspension coordinates in $S^n = (S^1)^{\wedge n}$.

Start with *symmetric sequences*. For two symmetric sequences A and B , set

$$(A \otimes B)_n = \bigvee_{p+q=n} (\Sigma_n)_+ \wedge_{\Sigma_p \times \Sigma_q} (A_p \wedge B_q). \quad (8.6)$$

For a subgroup $H \subseteq G$ and an H -space Z ,

$$\text{Ind}_H^G Z = G_+ \wedge_H Z$$

is obtained from pairs (g, z) modulo $(gh, z) \sim (g, hz)$.

The sphere sequence $\mathbb{S}_n = S^n$ is a commutative monoid for (8.6). Symmetric spectra are modules over this monoid. Their smash product is the relative tensor product

$$X \wedge_{\mathbb{S}} Y = \operatorname{coeq}(X \otimes \mathbb{S} \otimes Y \rightrightarrows X \otimes Y), \quad (8.7)$$

with unit $\mathbb{S}$.

Let evaluation at level n take values in based Σ_n -spaces. Its left adjoint F_n satisfies, above level n ,

$$(F_n K)_{n+p} \simeq (\Sigma_{n+p})_+ \wedge_{\Sigma_n \times \Sigma_p} (K \wedge S^p), \quad (8.8)$$

for a based Σ_n -space K .

In the stable model structure the fibrant objects are levelwise fibrant Ω -spectra. For cofibrations $i: A \rightarrow B$ and $j: K \rightarrow L$,

$$i \square j: (B \wedge K) \amalg_{A \wedge K} (A \wedge L) \longrightarrow B \wedge L \quad (8.9)$$

is a cofibration, acyclic when either input is acyclic. The derived smash is computed by $QX \wedge QY$ under the standard cofibrancy hypotheses.

Under the usual hypotheses, transfer the model structure to associative monoids and their modules. For strictly commutative monoids, there is a problem with symmetric powers at level zero. Usually use the positive stable model structure here.

8.4 Orthogonal spectra and Thom mapping spaces

For orthogonal spectra, replace the index n by a finite-dimensional real inner-product space V . Let $\mathcal{L}(V, W)$ be the space of linear isometric embeddings $V \hookrightarrow W$. The complement bundle over it is

$$\xi(V, W) = \{(f, w) \mid f \in \mathcal{L}(V, W), w \perp f(V)\},$$

whose fiber at f is $W - f(V)$. The enriched mapping object of the orthogonal indexing category is

$$\mathcal{O}(V, W) = \operatorname{Th}(\xi(V, W)). \quad (8.10)$$

Compose using isometries and orthogonal complements. Direct sum of inner-product spaces makes $\mathcal{O}$ symmetric monoidal.

For a vector bundle $E \rightarrow B$,

$$\operatorname{Th}(E) = D(E)/S(E),$$

the fiberwise one-point compactification with the infinity section collapsed. For $V = 0$, $\mathcal{O}(0, W) \cong S^W$.

Definition 8.2 (Orthogonal spectrum). An *orthogonal spectrum* is a continuous enriched functor $X: \mathcal{O} \rightarrow \mathbf{Top}_*$. Equivalently, it has based $\mathcal{O}(V)$ -spaces $X(V)$ and compatible structure maps

$$X(V) \wedge S^W \longrightarrow X(V \oplus W).$$

Day convolution gives the smash product. Monoidal Quillen equivalences compare orthogonal and symmetric spectra. There is also the model using S -modules.

8.5 The excisive-functor model

Reduced excisive functors on finite pointed CW spaces also model spectra. A spectrum E gives

$$K \longmapsto \Omega^\infty(E \wedge \Sigma^\infty K). \quad (8.11)$$

Excision gives stability. The suspension square

$$\begin{array}{ccc} K & \longrightarrow & * \\ \downarrow & & \downarrow \\ * & \longrightarrow & \Sigma K \end{array}$$

is a homotopy pushout. A reduced excisive functor gives $F(K) \simeq \Omega F(\Sigma K)$. Iterate to get compatible deloopings.

9 Γ -Spaces and Connective Spectra

9.1 Special and very special Γ -spaces

Let Γ be the category of finite pointed sets, and write

$$n_+ = \{0, 1, \dots, n\},$$

with basepoint 0. Here a Γ -space is a pointed functor $A: \Gamma \rightarrow \mathbf{Top}_*$. Authors who use the opposite category for Γ write this contravariantly. For each $1 \leq i \leq n$, the map $p_i: n_+ \rightarrow 1_+$ that sends i to 1 and all other elements to 0 induces a *Segal map*

$$A(n_+) \longrightarrow A(1_+)^n. \quad (9.1)$$

Definition 9.1 (Γ -space). A Γ -space A is *special* if $A(0_+) \simeq *$ and every Segal map (9.1) is a weak equivalence. It is *very special* if, in addition, the commutative monoid $\pi_0 A(1_+)$ induced by the fold map $2_+ \rightarrow 1_+$ is a group.

Special means $A(n_+) \simeq A(1_+)^n$; very special adds group-likeness.

Extend A from finite pointed sets to pointed spaces and evaluate it on spheres. This gives an associated spectrum:

$$\mathbf{B}A = (A(S^0), A(S^1), A(S^2), \dots). \quad (9.2)$$

The assembly maps are its structure maps. Very special Γ -spaces model connective spectra.

9.2 Smash product and model structures

Finite pointed sets form a symmetric monoidal category under smash product, with unit 1_+ . Day convolution therefore gives a smash product of Γ -spaces:

$$(A \wedge_\Gamma B)(n_+) = \int^{k_+, \ell_+ \in \Gamma} \underline{\Gamma}(k_+ \wedge \ell_+, n_+) \wedge A(k_+) \wedge B(\ell_+). \quad (9.3)$$

Each hom-set is based by the zero map, and the unit is $\underline{\Gamma}(1_+, -)$. Derived smash products require cofibrant inputs.

Localizing the objectwise model structure at the Segal and group-completion maps gives diagrams $\rightarrow$ homotopy-coherent monoids $\rightarrow$ grouplike monoids $\rightarrow$ connective spectra.

9.3 Algebraic K -theory

For a permutative category $\mathcal{C}$, the Γ -space construction and group completion give $K(\mathcal{C})$, with

$$\Omega^\infty K(\mathcal{C}) \simeq (BC)^{\text{gp}}$$

after deriving.

Examples:

1. For the symmetric monoidal groupoid of finite sets and bijections under disjoint union, group completion produces the sphere-spectrum infinite loop space. This is the content of the Barratt–Priddy–Quillen theorem.
2. For the maximal subgroupoid of finitely generated projective R -modules and isomorphisms under direct sum, the machine gives algebraic K -theory $K(R)$.
3. The topological groupoids of finite-dimensional complex, respectively real, vector spaces and linear isomorphisms give connective topological K -theory ku , respectively ko , after group completion.

10 Stable Model Categories and Spectral Enrichment

10.1 Stability as a Quillen condition

Let $\mathcal{M}$ be a pointed model category. In a simplicial or topological model category, tensor and cotensor with S^1 give a Quillen adjunction

$$\Sigma: \mathcal{M} \rightleftarrows \mathcal{M}: \Omega.$$

In general, define suspension by a homotopy cofiber and loops by a homotopy fiber.

Definition 10.1 (Stable model category). A pointed model category $\mathcal{M}$ is *stable* if the derived adjunction

$$\mathbf{L}\Sigma: \text{Ho}(\mathcal{M}) \rightleftarrows \text{Ho}(\mathcal{M}): \mathbf{R}\Omega$$

is an adjoint equivalence.

Examples include spectra, unbounded chain complexes, and module spectra. Finite homotopy pushouts and pullbacks agree.

10.2 SM7 and enrichment

In a simplicial model category, let $i: A \rightarrow B$ be a cofibration and $p: X \rightarrow Y$ a fibration. Axiom SM7 says that the induced map of simplicial mapping objects

$$\text{Map}(B, X) \longrightarrow \text{Map}(A, X) \times_{\text{Map}(A, Y)} \text{Map}(B, Y) \quad (10.1)$$

is a Kan fibration. It is an acyclic fibration when either i or p is a weak equivalence. By adjunction, this is equivalent to the pushout–product condition for tensoring $\mathcal{M}$ over simplicial sets.

For cofibrant A and fibrant X , $\text{Map}(A, X)$ is already a derived mapping space. Spectral model categories replace simplicial sets by spectra.

Definition 10.2 (Spectral model category). A *spectral model category* is a model category $\mathcal{M}$ enriched, tensored, and cotensored over a chosen symmetric monoidal model category of spectra, with adjunctions

$$\mathcal{M}(K \wedge X, Y) \cong \mathbf{Sp}(K, \underline{\mathrm{Map}}_{\mathcal{M}}(X, Y)) \cong \mathcal{M}(X, Y^K),$$

and satisfying the spectral version of SM7.

For cofibrant X and fibrant Y , the enriched mapping spectrum computes all stable mapping groups:

$$\pi_n \underline{\mathrm{Map}}_{\mathcal{M}}(X, Y) \cong \mathrm{Ho}(\mathcal{M})(\Sigma^n X, Y), \quad n \in \mathbb{Z}. \quad (10.2)$$

10.3 Spectral categories

Definition 10.3 (Spectral category). A *spectral category* $\mathcal{A}$ is a category enriched in spectra. For objects a, b it has a mapping spectrum $\mathcal{A}(a, b)$, unit maps $\mathbb{S} \rightarrow \mathcal{A}(a, a)$, and associative composition maps

$$\mathcal{A}(b, c) \wedge \mathcal{A}(a, b) \longrightarrow \mathcal{A}(a, c).$$

A one-object spectral category is a ring spectrum. Modules over a spectral category are enriched functors $\mathcal{A}^{\mathrm{op}} \rightarrow \mathbf{Sp}$.

Under the spectral Morita hypotheses, choose compact generators. Then $\mathcal{M}$ is modeled by modules over their full spectral subcategory.

The set $\pi_0 \mathcal{A}(a, b)$ is the morphism set in the homotopy category.

10.4 Spectra internal to a model category

Start with a pointed model category $\mathcal{M}$. If $\mathcal{M}$ is tensored over pointed simplicial sets, a symmetric spectrum in $\mathcal{M}$ consists of objects $X_n \in \mathcal{M}$ with Σ_n -actions and equivariant maps

$$S^1 \otimes X_n \longrightarrow X_{n+1}.$$

Localize the level model structure at the adjoint structure maps to get $\mathbf{Sp}^{\Sigma}(\mathcal{M})$ under the usual hypotheses.

Evaluation at zero has a suspension-spectrum left adjoint:

$$\Sigma_{\mathcal{M}}^{\infty}: \mathcal{M} \rightleftarrows \mathbf{Sp}^{\Sigma}(\mathcal{M}): \mathrm{Ev}_0. \quad (10.3)$$

If $\mathcal{M}$ is already stable, this is a Quillen equivalence under the usual hypotheses.

11 Stabilization in ∞ -Categories

11.1 Excisive functors as spectrum objects

Let $\mathcal{C}$ be an ∞ -category with finite limits and a terminal object. A functor

$$F: \mathcal{S}_*^{\mathrm{fin}} \longrightarrow \mathcal{C}$$

is *reduced* if $F(*)$ is terminal and *excisive* if it sends pushout squares of finite pointed spaces to pullback squares in $\mathcal{C}$.

Definition 11.1 (Stabilization). The ∞ -category of spectrum objects in $\mathcal{C}$ is

$$\mathbf{Sp}(\mathcal{C}) = \mathrm{Exc}_*(\mathcal{S}_*^{\mathrm{fin}}, \mathcal{C}), \quad (11.1)$$

the full subcategory of reduced excisive functors. When $\mathcal{C}$ is the ∞ -category $\mathcal{S}$ of spaces, we write simply $\mathbf{Sp}$.

Excision gives equivalences

$$F(S^n) \simeq \Omega F(S^{n+1}).$$

Equivalently, a spectrum object is a coherent sequence

$$(X_0, X_1, \dots), \quad X_n \xrightarrow{\simeq} \Omega X_{n+1}. \quad (11.2)$$

Under the usual presentability assumptions, there is an adjunction

$$\Sigma^\infty : \mathcal{C} \rightleftarrows \mathbf{Sp}(\mathcal{C}) : \Omega^\infty, \quad (11.3)$$

where Ω^∞ evaluates at S^0 . For an unpointed category, write Σ_+^∞ .

11.2 Symmetric monoidal stabilization and Day convolution

Day convolution gives the smash product on spectrum objects. For spaces it gives $(\mathbf{Sp}, \wedge, \mathbb{S})$.

If $\mathcal{C}^\otimes$ is presentably symmetric monoidal and tensor preserves colimits separately, its stabilization is symmetric monoidal, with

$$\mathrm{CAlg}(\mathbf{Sp}(\mathcal{C})), \quad \mathrm{Alg}(\mathbf{Sp}(\mathcal{C})), \quad \mathrm{Mod}_R(\mathbf{Sp}(\mathcal{C})).$$

The shifts of the unit represent stable homotopy groups:

$$\pi_n X \cong \mathrm{Ho}(\mathbf{Sp})(\Sigma^n \mathbb{S}, X).$$

Understanding / Intuition

The multiplication $R \wedge R \rightarrow R$ is not the addition. Addition on $\pi_ R$ comes from stability; multiplication gives the products $\pi_p R \otimes \pi_q R \rightarrow \pi_{p+q} R$.*

If $\mathcal{C}$ is pointed and presentable and $\mathcal{D}$ is stable and presentable, precomposition with Σ^∞ induces

$$\mathrm{Fun}^L(\mathbf{Sp}(\mathcal{C}), \mathcal{D}) \simeq \mathrm{Fun}^L(\mathcal{C}, \mathcal{D}).$$

This characterizes $\mathbf{Sp}(\mathcal{C})$ as the universal stable recipient [Lur17, Sections 1.4 and 4.8].

If $\mathcal{C}$ is already stable, the canonical functor $\mathbf{Sp}(\mathcal{C}) \rightarrow \mathcal{C}$ is an equivalence.

11.3 Stable ∞ -categories

Definition 11.2 (Stable ∞ -category). An ∞ -category $\mathcal{C}$ is *stable* if

1. it has a zero object;
2. it has all finite limits and finite colimits; and
3. a commutative square in $\mathcal{C}$ is a pushout if and only if it is a pullback.

For a map $f: X \rightarrow Y$, its cofiber and fiber are defined by the two squares

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \operatorname{cof}(f) \end{array} \quad \begin{array}{ccc} \operatorname{fib}(f) & \longrightarrow & X \\ \downarrow & & \downarrow f \\ 0 & \longrightarrow & Y. \end{array}$$

In a stable category:

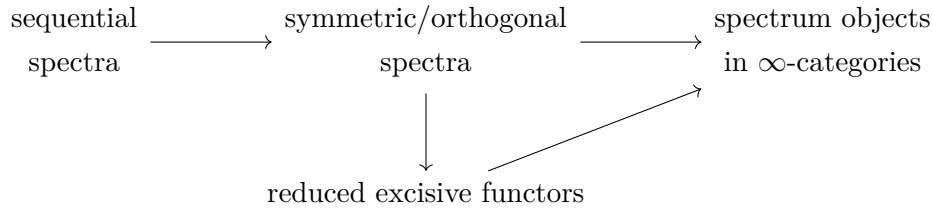
$$\operatorname{cof}(f) \simeq \Sigma \operatorname{fib}(f), \quad \operatorname{fib}(f) \simeq \Omega \operatorname{cof}(f).$$

In particular, Σ and Ω are inverse equivalences. Finite products and coproducts agree, giving biproducts $X \oplus Y$.

Every stable ∞ -category is canonically enriched in spectra. For $X, Y \in \mathcal{C}$, the mapping space is an infinite loop space. It extends to a mapping spectrum $\operatorname{map}_{\mathcal{C}}(X, Y)$ satisfying

$$\Omega^\infty \operatorname{map}_{\mathcal{C}}(X, Y) \simeq \operatorname{Map}_{\mathcal{C}}(X, Y), \quad \pi_n \operatorname{map}_{\mathcal{C}}(X, Y) \cong \operatorname{Ho}(\mathcal{C})(\Sigma^n X, Y). \quad (11.4)$$

A functor between stable ∞ -categories is *exact* if it preserves finite limits, equivalently finite colimits.



The arrows compare the derived homotopy theories, not point-set objects.

12 Triangulated Categories and t -Structures

12.1 The triangulated homotopy category

Let $\mathcal{C}$ be a stable ∞ -category. Its homotopy category $\operatorname{Ho}(\mathcal{C})$ is additive, suspension induces an autoequivalence [1], and every morphism $f: X \rightarrow Y$ determines a triangle

$$X \xrightarrow{f} Y \longrightarrow \operatorname{cof}(f) \longrightarrow \Sigma X. \quad (12.1)$$

Triangles isomorphic to those in (12.1) are distinguished and satisfy the triangulated axioms.

For any $Z \in \mathcal{C}$, applying the mapping-spectrum functor to (12.1) gives a fiber sequence of spectra and hence a long exact sequence

$$\cdots \rightarrow \operatorname{Ho}(\mathcal{C})(\Sigma X, Z) \rightarrow \operatorname{Ho}(\mathcal{C})(\operatorname{cof}(f), Z) \rightarrow \operatorname{Ho}(\mathcal{C})(Y, Z) \rightarrow \operatorname{Ho}(\mathcal{C})(X, Z) \rightarrow \cdots.$$

Mapping out of Z gives the covariant version.

12.2 t -Structures

Definition 12.1 (t -structure). A t -structure on a stable ∞ -category $\mathcal{C}$ is a pair of full subcategories $(\mathcal{C}_{\geq 0}, \mathcal{C}_{\leq 0})$ such that, writing

$$\mathcal{C}_{\geq n} = \Sigma^n \mathcal{C}_{\geq 0}, \quad \mathcal{C}_{\leq n} = \Sigma^n \mathcal{C}_{\leq 0},$$

the following conditions hold.

1. $\mathcal{C}_{\geq 1} \subseteq \mathcal{C}_{\geq 0}$ and $\mathcal{C}_{\leq 0} \subseteq \mathcal{C}_{\leq 1}$.
2. If $X \in \mathcal{C}_{\geq 0}$ and $Y \in \mathcal{C}_{\leq -1}$, then $\mathrm{Map}_{\mathcal{C}}(X, Y)$ is contractible.
3. Every $X \in \mathcal{C}$ fits into a fiber/cofiber sequence

$$\tau_{\geq 0} X \longrightarrow X \longrightarrow \tau_{\leq -1} X \quad (12.2)$$

with $\tau_{\geq 0} X \in \mathcal{C}_{\geq 0}$ and $\tau_{\leq -1} X \in \mathcal{C}_{\leq -1}$.

The inclusion $\mathcal{C}_{\geq 0} \hookrightarrow \mathcal{C}$ has right adjoint $\tau_{\geq 0}$, while $\mathcal{C}_{\leq -1} \hookrightarrow \mathcal{C}$ has left adjoint $\tau_{\leq -1}$. Shifting gives all truncation functors $\tau_{\geq n}$ and $\tau_{\leq n}$. The *heart*

$$\mathcal{C}^{\heartsuit} = \mathcal{C}_{\geq 0} \cap \mathcal{C}_{\leq 0}$$

is an abelian category.

The standard t -structure on spectra is

$$\mathbf{Sp}_{\geq 0} = \{X \mid \pi_n X = 0 \text{ for } n < 0\}, \quad \mathbf{Sp}_{\leq 0} = \{X \mid \pi_n X = 0 \text{ for } n > 0\}.$$

Its heart is equivalent to $\mathbf{Ab}$: an abelian group A corresponds to the Eilenberg–Mac Lane spectrum HA . The truncation tower is the Postnikov tower

$$\cdots \longrightarrow \tau_{\leq n} X \longrightarrow \tau_{\leq n-1} X \longrightarrow \cdots,$$

whose successive layers are $\Sigma^n H \pi_n X$.

Example 12.2 (Chain complexes). For chain complexes of R -modules, the standard t -structure is given by homology in degrees ≥ 0 and ≤ 0 , and its heart is R -modules.

A t -structure sorts objects by homological degree in a stable category. It is not a model structure.

PART III

Day 3: Rings, Modules, and Coherent Multiplication

13 Ring Spectra and Their Algebra

13.1 Monoid objects in spectra

Use one of the symmetric monoidal point-set models $(\mathbf{Sp}, \wedge, \mathbb{S})$ from Day 2. The unit is the sphere spectrum $\mathbb{S}$. Compare

$$(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z}) \quad \rightsquigarrow \quad (\mathbf{Sp}, \wedge, \mathbb{S}).$$

An ordinary ring is a monoid object in abelian groups. A ring spectrum is a monoid object in spectra.

Definition 13.1 (Ring spectrum). An *associative ring spectrum* is a spectrum R with multiplication and unit

$$\mu: R \wedge R \longrightarrow R, \quad \eta: \mathbb{S} \longrightarrow R,$$

such that the diagrams

$$\begin{array}{ccc} (R \wedge R) \wedge R & \xrightarrow{\alpha} & R \wedge (R \wedge R) \\ \mu \wedge \text{id} \downarrow & & \downarrow \text{id} \wedge \mu \\ R \wedge R & \xrightarrow{\mu} & R \end{array} \quad \begin{array}{ccccc} \mathbb{S} \wedge R & \xrightarrow{\eta \wedge \text{id}} & R \wedge R & \xleftarrow{\text{id} \wedge \eta} & R \wedge \mathbb{S} \\ & \searrow \lambda_R & \downarrow \mu & \swarrow \rho_R & \\ & & R & & \end{array}$$

commute. Here α, λ , and ρ are the coherence isomorphisms of the monoidal category. It is *commutative* if in addition

$$\begin{array}{ccc} R \wedge R & \xrightarrow{\tau} & R \wedge R \\ & \searrow \mu & \downarrow \mu \\ & & R \end{array}$$

commutes, where τ interchanges the two factors.

Monoidal Quillen equivalences compare symmetric, orthogonal, and S -module models. In the underlying theory, these are E_1 - and commutative algebra objects in spectra [Lur17].

Example 13.2. The unit $\mathbb{S}$ is a commutative ring spectrum. Every ordinary ring A has an Eilenberg–Mac Lane spectrum HA with

$$\pi_k(HA) \cong \begin{cases} A, & k = 0, \\ 0, & k \neq 0. \end{cases}$$

If A is unital, HA is E_1 ; if A is commutative, it is E_∞ . Other examples include KO , KU , and multiplicative Thom spectra.

13.2 Modules, algebras, and commutative algebras

Fix an associative ring spectrum R .

Definition 13.3 (R -module spectrum). A *left R -module spectrum* is a spectrum M with an action

$$a: R \wedge M \longrightarrow M$$

such that

$$\begin{array}{ccc} (R \wedge R) \wedge M & \xrightarrow{\alpha} & R \wedge (R \wedge M) \\ \mu \wedge \text{id} \downarrow & & \downarrow \text{id} \wedge a \\ R \wedge M & \xrightarrow{a} & M \end{array}$$

commutes and $a(\eta \wedge \text{id})$ is the unit isomorphism $\mathbb{S} \wedge M \cong M$. Right modules and (R, S) -bimodules are defined similarly.

For commutative R , Mod_R has relative smash product $\wedge_R$ and unit R . This is symmetric monoidal. Can take algebra objects here too.

Definition 13.4 (R -algebra). Let R be commutative. An *associative R -algebra* is a monoid object in $(\text{Mod}_R, \wedge_R, R)$; explicitly it has maps

$$A \wedge_R A \longrightarrow A, \quad R \longrightarrow A.$$

A *commutative R -algebra* is a commutative monoid object in Mod_R . We write

$$\text{Alg}_R = \text{Alg}(\text{Mod}_R), \quad \text{CAlg}_R = \text{CAlg}(\text{Mod}_R).$$

The forgetful functors are

$$\text{CAlg}_R \xrightarrow{U} \text{Alg}_R \xrightarrow{U} \text{Mod}_R \xrightarrow{U} \mathbf{Sp},$$

The arrows forget structure. Set $R = \mathbb{S}$ for the absolute versions.

There is a free-forgetful adjunction

$$\mathbb{P}_R: \text{Mod}_R \rightleftarrows \text{CAlg}_R: U, \quad \mathbb{P}_R \dashv U,$$

where U forgets the commutative multiplication and $\mathbb{P}_R$ is the free commutative R -algebra functor. In a suitable point-set category,

$$\mathbb{P}_R(M) = \bigvee_{n \geq 0} M^{\wedge_R n} / \Sigma_n, \quad M^{\wedge_R 0} = R.$$

Derived coinvariants are written $M_{h\Sigma_n}^{\wedge_R n}$.

A map

$$\mathbb{P}_R(M) \longrightarrow A \quad \text{in } \text{CAlg}_R$$

is equivalent to an R -module map

$$M \longrightarrow U(A).$$

13.3 Transferred and positive model structures

Transfer the stable model structure across the free–forgetful adjunction: a map of algebras is a weak equivalence or fibration when its underlying map of modules is one; cofibrations are determined by the lifting property.

For modules and associative algebras, transfer works under the monoid-axiom and smallness hypotheses. Commutative algebras need more assumptions. Usually use the positive stable model structure.

The monoid axiom controls cell complexes built from maps

$$j \wedge X,$$

with j acyclic.

Positive cofibrations: isomorphisms in level zero, ordinary cofibrations above it. Same stable weak equivalences. Symmetric powers now work for strict commutative ring spectra.

The Lewis obstruction: cannot have both a cofibrant sphere and the expected model structure on strictly commutative monoids. Positive model structures avoid this. Stable weak equivalences stay the same [MMSS01].

13.4 Tensor, cotensor, and enriched mapping spectra

The module category is enriched, tensored, and cotensored over spectra. For $K \in \mathbf{Sp}$ and $M, N \in \mathbf{Mod}_R$, it has a tensor $K \wedge M$, a cotensor $F(K, N)$, and a mapping spectrum $\underline{\mathrm{Map}}_R(M, N)$ characterized by natural isomorphisms

$$\begin{aligned} \mathbf{Mod}_R(K \wedge M, N) &\cong \mathbf{Sp}(K, \underline{\mathrm{Map}}_R(M, N)), \\ \mathbf{Mod}_R(M, F(K, N)) &\cong \mathbf{Sp}(K, \underline{\mathrm{Map}}_R(M, N)). \end{aligned}$$

The composition maps

$$\underline{\mathrm{Map}}_R(N, P) \wedge \underline{\mathrm{Map}}_R(M, N) \longrightarrow \underline{\mathrm{Map}}_R(M, P)$$

make $\mathbf{Mod}_R$ a spectral category.

If M is a right R -module and N a left R -module, then

$$M \wedge_R^{\mathbf{L}} N \simeq Q^{\mathrm{flat}} M \wedge_R Q^{\mathrm{flat}} N$$

under the usual flatness hypotheses. There is then a spectral sequence

$$\mathrm{Tor}_{p,q}^{\pi_* R}(\pi_* M, \pi_* N) \Longrightarrow \pi_{p+q}(M \wedge_R^{\mathbf{L}} N).$$

14 Morita Theory for Ring Spectra

14.1 Classical Morita theory and bimodules

Two nonisomorphic rings can have equivalent module categories.

Definition 14.1 (Morita equivalence). Rings R and S are *Morita equivalent* if there is an additive equivalence

$$R\text{-Mod} \simeq S\text{-Mod}.$$

In derived Morita theory one instead asks for an equivalence of derived module categories, preserving the relevant enrichment and colimits.

With the convention that an (S, R) -bimodule ${}_S P_R$ points from R to S , it induces

$$P \otimes_R - : R\text{-Mod} \longrightarrow S\text{-Mod}.$$

If ${}_T Q_S$ points from S to T , composition is tensor product:

$$R \xrightarrow{{}_S P_R} S \xrightarrow{{}_T Q_S} T \quad \rightsquigarrow \quad {}_T(Q \otimes_S P)_R.$$

The identity at R is ${}_R R_R$; composition is associative up to the usual isomorphism.

For ring spectra, tensor products are replaced by derived relative smash products. An (S, R) -bimodule spectrum P gives

$$P \wedge_R^{\mathbf{L}} - : \mathrm{Ho}(\mathrm{Mod}_R) \longrightarrow \mathrm{Ho}(\mathrm{Mod}_S),$$

and invertible bimodules give derived Morita equivalences.

14.2 The matrix-ring test case

For a unital ring R and $n \geq 1$, we have

$$R \simeq_{\mathrm{Mor}} M_n(R).$$

Let $P = R^n$ be the column module, viewed as an $(M_n(R), R)$ -bimodule, and let

$$P^\vee = \mathrm{Hom}_R(P, R)$$

where Hom_R means homomorphisms of right R -modules. This is the dual row module, an $(R, M_n(R))$ -bimodule. Then

$$P \otimes_R P^\vee \cong \mathrm{End}_R(P) \cong M_n(R), \quad P^\vee \otimes_{M_n(R)} P \cong R.$$

The functors from P and $P^\vee$ are inverse equivalences of module categories. For spectra, replace R^n by a finite wedge of copies of a module. Replace tensor products by derived smash products.

Morita equivalence keeps the module theory; matrix size can change. Algebraic K -theory is Morita invariant.

14.3 A stable theory with one compact generator

Let $\mathcal{M}$ be a stable homotopy theory and $G \in \mathcal{M}$. Its derived endomorphism object is the ring spectrum

$$R = \mathbf{R}\underline{\mathrm{End}}_{\mathcal{M}}(G) = \mathbf{R}\underline{\mathrm{Map}}_{\mathcal{M}}(G, G).$$

There is a derived adjunction

$$\mathrm{Mod}_R^{\mathrm{right}} \begin{array}{c} \xrightarrow{-\wedge_R^{\mathbf{L}} G} \\ \xleftarrow{\mathbf{R}\underline{\mathrm{Map}}(G, -)} \end{array} \mathcal{M}.$$

Here $\mathrm{Mod}_R^{\mathrm{right}}$ means right R -modules. Precomposition gives $\mathbf{R}\underline{\mathrm{Map}}(G, X)$ a right R -action. Evaluation gives G a left R -action. These are the sides needed for the relative smash.

Theorem 14.2 (Schwede–Shipley Morita theorem, one-generator form). *Let $\mathcal{M}$ be a simplicial, cofibrantly generated, proper, stable model category whose homotopy category admits a compact generator G . After choosing a cofibrant–fibrant representative, $\mathcal{M}$ is connected by a zigzag of simplicial Quillen equivalences to right modules over the derived endomorphism ring spectrum of G :*

$$\mathcal{M} \simeq_{\mathbf{Q}} \mathrm{Mod}_{\mathbf{R}\mathrm{End}(G)}^{\mathrm{right}}.$$

In particular, the stable homotopy theory is recovered from G together with all of its stable self-maps and their higher homotopies [SS03].

Here compactness means

$$[G, \bigvee_i X_i] \cong \bigoplus_i [G, X_i],$$

and generation means $[\Sigma^n G, X] = 0$ for all n implies $X \simeq 0$.

14.4 Several generators: ringoids and spectral categories

For a set $\mathcal{G} = \{G_i\}_{i \in I}$ of compact generators, use a many-object endomorphism algebra.

Definition 14.3 (Spectral endomorphism category). Choose cofibrant–fibrant representatives in a spectral model or a spectral replacement of $\mathcal{M}$. The *spectral endomorphism category* $\mathcal{O}_{\mathcal{G}}$ has objects the chosen generators G_i and mapping spectra

$$\mathcal{O}_{\mathcal{G}}(G_i, G_j) = \mathbf{R}\underline{\mathrm{Map}}_{\mathcal{M}}(G_i, G_j).$$

The spectral presentation has strict enriched composition

$$\mathcal{O}_{\mathcal{G}}(G_j, G_k) \wedge \mathcal{O}_{\mathcal{G}}(G_i, G_j) \longrightarrow \mathcal{O}_{\mathcal{G}}(G_i, G_k).$$

A right $\mathcal{O}_{\mathcal{G}}$ -module is a spectral functor

$$\mathcal{O}_{\mathcal{G}}^{\mathrm{op}} \longrightarrow \mathbf{Sp}.$$

For compact generators,

$$\text{one generator} \longrightarrow \text{many generators}$$

corresponds to

$$\text{ring spectrum} \longrightarrow \text{spectral category}.$$

The many-generator theorem models $\mathcal{M}$ by modules over $\mathcal{O}_{\mathcal{G}}$.

15 Bar Constructions and Derived Tensor Products

Construction 15.1 (Two-sided bar construction). Let M be a right R -module and N a left R -module. The simplicial spectrum $B_{\bullet}(M, R, N)$ is defined in degree q by

$$B_q(M, R, N) = M \wedge R^{\wedge q} \wedge N.$$

The two outer face maps use the actions on M and N ; an inner face map multiplies two adjacent copies of R ; and each degeneracy inserts the unit $\mathbb{S} \rightarrow R$. Its realization is

$$B(M, R, N) = |B_{\bullet}(M, R, N)|.$$

There is a natural comparison

$$B(M, R, N) \longrightarrow M \wedge_R N.$$

Under the usual cofibrancy hypotheses, the bar construction models the derived relative smash product:

$$B(M, R, N) \simeq M \wedge_R^{\mathbf{L}} N.$$

In low degrees the simplicial object is

$$M \wedge N \quad \rightrightarrows \quad M \wedge R \wedge N \quad \rightrightarrows \quad M \wedge R \wedge R \wedge N \quad \cdots.$$

Realization imposes the R -balancing relation by a derived coequalizer.

Definition 15.2 (Cyclic bar construction). For an associative ring spectrum R , its cyclic bar object has

$$B_q^{\text{cy}}(R) = R^{\wedge(q+1)}.$$

The ordinary faces multiply adjacent factors, the last face multiplies the last and first factors after a cyclic permutation, and the degeneracies insert the unit. The cyclic symmetry gives the realization a circle action. After deriving the construction,

$$\text{THH}(R) \simeq |B_{\bullet}^{\text{cy}}(R)|.$$

Coefficients in an R -bimodule give $B^{\text{cy}}(R; M)$ when the required cyclic structure is present.

The circle action gives fixed-point and Tate constructions. For topological cyclic homology, also need cyclotomic Frobenius maps.

16 Operads and Coherent Algebra

16.1 Loop-space recognition

For which spaces X is there an equivalence

$$X \simeq \Omega^n Y$$

for some Y ? Loop concatenation gives ΩY a multiplication. Recognition requires a grouplike coherent E_n -structure.

Example 16.1 (Unit octonions). Unit octonions: S^7 has a multiplication, a unit, and inverses, but no associativity for this multiplication. An H -space multiplication alone is not enough for loop-space recognition.

16.2 Endomorphism operads

Let $(\mathcal{V}, \otimes, \mathbf{1})$ be symmetric monoidal with internal Homs. The endomorphism operad records multilinear operations on X .

Definition 16.2 (Endomorphism operad). The *endomorphism operad* of X is the symmetric sequence

$$\text{End}_X(n) = \underline{\text{Hom}}_{\mathcal{V}}(X^{\otimes n}, X), \quad n \geq 0.$$

The group Σ_n acts by permuting the n inputs. Operadic composition is substitution:

$$\begin{aligned} \text{End}_X(k) \otimes \text{End}_X(n_1) \otimes \cdots \otimes \text{End}_X(n_k) \\ \longrightarrow \text{End}_X(n_1 + \cdots + n_k), \end{aligned}$$

and the identity map of X gives the unit $\mathbf{1} \rightarrow \text{End}_X(1)$.

For spaces, $\text{End}_X(n) = \text{Map}(X^n, X)$; for spectra it is $F(X^{\wedge n}, X)$.

16.3 Symmetric sequences and the substitution product

Definition 16.3 (Symmetric sequence). A *symmetric sequence* in a cocomplete symmetric monoidal category $\mathcal{V}$ is a family

$$A(0), A(1), A(2), \dots$$

in which $A(n)$ carries a right action of Σ_n . A morphism of symmetric sequences is a levelwise equivariant map.

Symmetric sequences have a substitution, or composition, product. Up to the standard choices of left and right actions, it is built from the terms

$$(A \circ B)(n) = \coprod_{k \geq 0} A(k) \otimes_{\Sigma_k} \left[\coprod_{n_1 + \dots + n_k = n} (B(n_1) \otimes \dots \otimes B(n_k)) \otimes_{\Sigma_{n_1} \times \dots \times \Sigma_{n_k}} \Sigma_n \right].$$

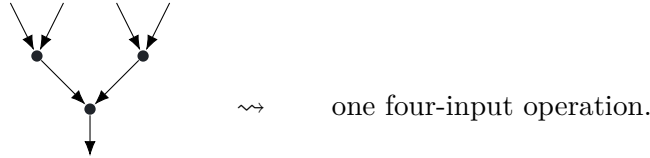
A term inserts operations of arities $n_1, \dots, n_k$ into a k -ary operation.

Definition 16.4 (Operad). An *operad* $\mathcal{O}$ is a monoid for the substitution product of symmetric sequences. Equivalently, it consists of objects $\mathcal{O}(n)$, Σ_n -actions, a unit $\mathbf{1} \rightarrow \mathcal{O}(1)$, and equivariant composition maps

$$\gamma: \mathcal{O}(k) \otimes \mathcal{O}(n_1) \otimes \dots \otimes \mathcal{O}(n_k) \longrightarrow \mathcal{O}(n_1 + \dots + n_k),$$

satisfying associativity and unit axioms.

Composition as a rooted tree:



Associativity: collapsing in stages or all at once gives the same result.

16.4 Algebras over an operad

Definition 16.5 ($\mathcal{O}$ -algebra). An *algebra over $\mathcal{O}$* is an object $X \in \mathcal{V}$ together with a map of operads

$$\mathcal{O} \longrightarrow \text{End}_X.$$

Equivalently, it has compatible structure maps

$$\mathcal{O}(n) \otimes_{\Sigma_n} X^{\otimes n} \longrightarrow X$$

for every $n \geq 0$.

17 Examples of Operads and Recognition

17.1 Strict associative and commutative operads

Example 17.1 (The commutative operad). Let

$$\text{Comm}(n) = \mathbf{1}$$

with trivial Σ_n -action and composition induced by the unit isomorphisms $\mathbf{1}^{\otimes k} \cong \mathbf{1}$. A Comm-algebra is a strictly commutative monoid in $\mathcal{V}$. Indeed, equivariance of the binary operation forces

$$\mu\tau = \mu.$$

For $\mathcal{V} = \mathbf{Sp}$ this gives a strict point-set presentation of a commutative ring spectrum when the ambient model structure admits it.

Example 17.2 (The associative operad). Take the copower

$$\mathrm{Ass}(n) = \coprod_{\sigma \in \Sigma_n} \mathbf{1}$$

with its regular action. In spaces this is the discrete object Σ_n . Its algebras are strictly associative unital monoids.

17.2 A_n , A_∞ , and little cubes

Stasheff's associahedra parametrize the coherences for A_n - and A_∞ -spaces.

Definition 17.3 (Little n -cubes operad). Let $\mathcal{C}_n(k)$ be the space of ordered k -tuples of pairwise interior-disjoint rectilinear embeddings

$$(0, 1)^n \hookrightarrow (0, 1)^n.$$

The group Σ_k relabels the small cubes. Operadic composition inserts a configuration of little cubes into each cube of another configuration. The resulting operad is the *little n -cubes operad*. An E_n -operad is an operad weakly equivalent, in an appropriate sense, to $\mathcal{C}_n$; an E_n -algebra is an algebra over such an operad.

For $n = 1$ this is coherent associativity; increasing n adds commutativity. An E_∞ -operad has contractible operation spaces with free symmetric group actions.

An n -fold loop space $\Omega^n Y$ carries an action of $\mathcal{C}_n$ by rescaling loops into the little cubes.

Theorem 17.4 (May recognition principle, schematic form). *Under the standard well-based/CW and Σ -free or cofibrant-operad hypotheses, a grouplike algebra X over a little n -cubes operad admits an n -fold delooping: there is a space $B^n X$ and a weak equivalence*

$$X \simeq \Omega^n B^n X.$$

Conversely, n -fold loop spaces naturally carry E_n -algebra structure. For $n = \infty$, a grouplike E_∞ -space is the zeroth space of a connective spectrum [May72].

Here *grouplike* means that $\pi_0 X$ is a group.

18 Γ -Spaces and Infinite Loop Spaces

18.1 Special Γ -spaces

Let

$$n_+ = \{0, 1, \dots, n\},$$

with basepoint 0, and let Fin_* denote the category of finite pointed sets and based maps.

Convention 18.1 (Variance). We write a Γ -space as a based functor

$$X: \mathbf{Fin}_* \longrightarrow \mathbf{Top}_*.$$

Equivalently, with $\Gamma = \mathbf{Fin}_*^{\text{op}}$ this is a functor $\Gamma^{\text{op}} \rightarrow \mathbf{Top}_*$.

For $1 \leq i \leq n$, let

$$p_i: n_+ \longrightarrow 1_+$$

send i to 1 and all other nonbasepoint elements to 0. These maps induce the Segal map

$$\varphi_n: X(n_+) \longrightarrow X(1_+)^n.$$

Definition 18.2 (Special and very special Γ -spaces). A Γ -space X is *special* if $X(0_+) \simeq *$ and every Segal map φ_n is a weak equivalence. It is *very special* if it is special and the commutative monoid $\pi_0 X(1_+)$ is a group. Equivalently, one may impose the usual shear-map condition.

The fold map

$$\nabla: 2_+ \longrightarrow 1_+$$

gives a multiplication after inverting the Segal equivalence:

$$X(1_+) \times X(1_+) \simeq X(2_+) \xrightarrow{X(\nabla)} X(1_+).$$

The very special condition is group-likeness.

18.2 The Segal machine

The Segal machine evaluates the extension of X on spheres:

$$E_n \simeq X(S^n)$$

and structure maps

$$\Sigma E_n \longrightarrow E_{n+1}.$$

For a fibrant very special Γ -space, the adjoint maps

$$E_n \longrightarrow \Omega E_{n+1}$$

are weak equivalences. Then $E = (E_0, E_1, \dots)$ is an Ω -spectrum, and

$$X(1_+) \simeq \Omega^\infty E.$$

Conversely, very special Γ -spaces model connective spectra.

The May–Thomason theorem compares machines satisfying suitable axioms.

19 Rectification and Its Limits

The Moore loop space is, schematically,

$$\Omega^M X = \{(\ell, \gamma) \mid \ell \geq 0, \gamma: [0, \ell] \rightarrow X, \gamma(0) = \gamma(\ell) = *\}.$$

For Moore loops, concatenate and add the lengths. Keeping the lengths makes concatenation strictly associative:

$$((\ell_1, \gamma_1) \cdot (\ell_2, \gamma_2)) \cdot (\ell_3, \gamma_3) = (\ell_1, \gamma_1) \cdot ((\ell_2, \gamma_2) \cdot (\ell_3, \gamma_3)).$$

With the compactly generated topology, this has the homotopy type of the ordinary loop space for well-pointed X .

More generally, a weak equivalence of operads

$$\mathcal{O} \longrightarrow \mathcal{P}$$

gives extension and restriction on algebras. Under the usual admissibility and cofibrancy hypotheses it is a Quillen equivalence.

Warning 19.1 (Strictification is contextual). Strictification depends on the operads and model structure. Moore loops fix this associativity problem. This does not settle commutative rectification.

20 Eilenberg–Mac Lane Spectra and Derived Algebra

20.1 From chain complexes to module spectra

For an ordinary ring A , the Eilenberg–Mac Lane construction gives a ring spectrum HA and sends chain complexes of A -modules to HA -module spectra. The connective case begins with the Dold–Kan correspondence

$$N : s\mathrm{Mod}_A \rightleftarrows \mathbf{Ch}_{\geq 0}(A) : \Gamma,$$

This is an equivalence between simplicial A -modules and nonnegative chain complexes. For the derived category, stabilize and use suitable unbounded models.

Theorem 20.1 (Modules over HA). *For an ordinary ring A , there is a chain of Quillen equivalences between an appropriate model category of unbounded chain complexes of A -modules and the model category of HA -module spectra. In particular,*

$$\mathrm{Ho}(HA\text{-Mod}) \simeq D(A),$$

and the equivalence is compatible with the stable homotopy-theoretic structure.

Theorem 20.2 (Shipley). *The homotopy theory of associative $H\mathbb{Z}$ -algebra spectra is equivalent, through a chain of Quillen equivalences, to the homotopy theory of associative differential graded $\mathbb{Z}$ -algebras.*

Strictly commutative comparisons require extra hypotheses, especially in positive characteristic.

20.2 Algebraic and nonalgebraic stable homotopy theories

Definition 20.3 (Algebraic stable homotopy theory). Informally, a stable model category or stable ∞ -category is *algebraic* if it is modeled by differential graded or otherwise chain-complex-enriched data. In standard examples, its mapping spectra are modules over $H\mathbb{Z}$ and are described by chain complexes. A stable theory without such an algebraic model is called *topological* or *nonalgebraic*.

Derived categories $D(A)$ are algebraic. Spectra are not: the endomorphism spectrum of the unit is

$$\underline{\mathrm{End}}_{\mathbf{Sp}}(\mathbb{S}) \simeq \mathbb{S},$$

which is not an $H\mathbb{Z}$ -module compatible with its unit.

Graded homotopy groups are not enough to determine a ring spectrum. Also need higher operations, k -invariants, and mapping spectra.

PART IV

Day 4: Rings and Modules, Part II

Convention: colimits, tensor products, and mapping objects are derived, unless a point-set model is specified.

21 Thom Spaces and the Stable J -Homomorphism

21.1 Fiberwise compactification

Let $p: E \rightarrow X$ be a real vector bundle of rank n . Let $E^+ \rightarrow X$ be its fiberwise one-point compactification, with section at infinity

$$s_\infty: X \longrightarrow E^+,$$

and the fiber over x is

$$(E_x)^+ \cong (\mathbb{R}^n)^+ \cong S^n.$$

Definition 21.1 (Thom space). The *Thom space* of E is the quotient

$$\mathrm{Th}(E) = E^+ / s_\infty(X).$$

Equivalently, after choosing a fiberwise inner product, it is the disk bundle modulo its sphere bundle:

$$\mathrm{Th}(E) \cong D(E) / S(E).$$

For the trivial bundle $\varepsilon^n = X \times \mathbb{R}^n$,

$$\mathrm{Th}(\varepsilon^n) \cong X_+ \wedge S^n \cong \Sigma^n X_+.$$

If $E \rightarrow X$ and $F \rightarrow Y$, then

$$\mathrm{Th}(E \boxplus F) \cong \mathrm{Th}(E) \wedge \mathrm{Th}(F),$$

where $E \boxplus F$ is the external direct sum over $X \times Y$.

A rank- n vector bundle is classified by a map

$$c_E: X \longrightarrow BO(n),$$

with $E \cong c_E^* \gamma_n$ for the universal bundle γ_n . The standard inclusions come from

$$\gamma_n \oplus \varepsilon^1 \longrightarrow \gamma_{n+1}, \quad BO(n) \longrightarrow BO(n+1).$$

Applying the Thom construction gives structure maps

$$\Sigma \mathrm{Th}(\gamma_n) \cong \mathrm{Th}(\gamma_n \oplus \varepsilon^1) \longrightarrow \mathrm{Th}(\gamma_{n+1}).$$

These spaces form MO ; analogous classifying spaces give MSO , $MSpin$, and MU .

21.2 Spherical fibrations and stabilization

Choose a model

$$G(n) = \mathrm{hAut}_*(S^n)$$

for based self-homotopy equivalences of S^n . One-point compactification gives

$$O(n) \longrightarrow G(n)$$

and hence a map

$$BO(n) \longrightarrow BG(n).$$

Add a trivial line on both sides, then take homotopy colimits. This gives the stable J -homomorphism:

$$BO \xrightarrow{J} BGL_1(\mathbb{S}).$$

Here $GL_1(\mathbb{S})$ is the unit space of the sphere spectrum. Keep the higher homotopy too.

Suppose a virtual bundle is represented stably by

$$\xi: X \longrightarrow BO \times \mathbb{Z}.$$

Compose with J to get a family of sphere lines. Its colimit is the Thom spectrum.

22 Units, Lines, and Generalized Thom Spectra

22.1 Sphere lines, R -lines, and the Thom colimit

An $\mathbb{S}$ -line is a spectrum equivalent to the sphere spectrum $\mathbb{S}$. The maximal ∞ -groupoid of such spectra and their equivalences is

$$\mathbb{S}\text{-line} \simeq B \mathrm{Aut}_{\mathbf{Sp}}(\mathbb{S}) \simeq BGL_1(\mathbb{S}).$$

More generally, let R be an E_∞ ring spectrum. An R -module L is an R -line if it is equivalent to R as an R -module. Let $R\text{-line}$ denote the maximal ∞ -groupoid of such modules and equivalences. The automorphism space of the line R is

$$\mathrm{Aut}_R(R) \simeq GL_1(R),$$

so there is an equivalence

$$R\text{-line} \simeq BGL_1(R).$$

This is the unit component of the full Picard space $\mathrm{Pic}(R)$.

Let

$$f: X \longrightarrow R\text{-line} \simeq BGL_1(R)$$

classify a local system of free rank-one R -modules on X .

Definition 22.1 (Generalized Thom spectrum). Regard f as a functor $X \rightarrow \mathrm{Mod}_R$. Its R -module *Thom spectrum* is

$$M_R f = \mathrm{colim}(X \xrightarrow{f} R\text{-line} \hookrightarrow \mathrm{Mod}_R).$$

When $R = \mathbb{S}$ and f is the composite of a stable vector bundle with J , this recovers the classical Thom spectrum. When f is nullhomotopic, the local system is constant, and

$$M_R f \simeq R \wedge \Sigma_+^\infty X.$$

22.2 Units and the assembly adjunction

The infinite loop space $\Omega^\infty R$ is an E_∞ -monoid. Its components form the multiplicative monoid $\pi_0 R$. Take the union of the unit components:

$$GL_1(R) \subseteq \Omega^\infty R, \quad \pi_0 GL_1(R) = (\pi_0 R)^\times.$$

It determines a connective spectrum $gl_1(R)$ with

$$\Omega^\infty gl_1(R) \simeq GL_1(R).$$

For $i > 0$,

$$\pi_i gl_1(R) \cong \pi_i R \quad (i > 0),$$

For grouplike E_∞ -spaces there is an adjunction

$$\Sigma_+^\infty : \mathbf{CGrp}(\mathbf{Top}) \rightleftarrows \mathbf{CAlg}(\mathbf{Sp}) : GL_1.$$

The counit at R is the assembly map

$$\Sigma_+^\infty GL_1(R) \longrightarrow R.$$

For an ordinary ring A , compare the homomorphism

$$\mathbb{Z}[A^\times] \longrightarrow A$$

from the group ring of its units.

22.3 Functoriality of the Thom construction

Functoriality of the Thom construction: a diagram over the space of lines

$$\begin{array}{ccc} X & \xrightarrow{\quad u \quad} & Y \\ & \searrow f \quad \quad \swarrow g & \\ & R\text{-line} & \end{array}$$

with a specified homotopy $f \simeq g \circ u$ induces

$$M_R f \longrightarrow M_R g.$$

22.4 Multiplicative Thom spectra

Given $f: X \rightarrow R\text{-line}$ and $g: Y \rightarrow R\text{-line}$, their external product $f \boxtimes g$ is classified by

$$X \times Y \xrightarrow{f \times g} R\text{-line} \times R\text{-line} \xrightarrow{\otimes_R} R\text{-line}.$$

The Thom functor is symmetric monoidal:

$$M_R(f \boxtimes g) \simeq M_R f \wedge_R M_R g.$$

If X is E_n -monoidal and f is E_n -monoidal, then $M_R f$ is an E_n R -algebra. An E_∞ -monoidal map gives a commutative R -algebra.

23 Topological Hochschild Homology

23.1 Simplicial and cyclic objects

For a simplicial object $X_\bullet$, realization is

$$|X_\bullet| = \operatorname{colim}_{[n] \in \Delta^{\text{op}}} X_n.$$

For the cyclic category, add a cyclic operator to Δ : one t_n in each degree. For a suitably presented associative R -algebra, the cyclic bar object is

$$B_n^{\text{cy}}(A/R) = A^{\wedge_R(n+1)}, \quad n \geq 0.$$

Faces multiply adjacent factors, degeneracies insert the unit, and t_n rotates the factors.

Definition 23.1 (Topological Hochschild homology). For an E_1 R -algebra A , its relative topological Hochschild homology is the realization

$$\operatorname{THH}^R(A) = |B_\bullet^{\text{cy}}(A/R)|.$$

In the absolute case $R = \mathbb{S}$, write $\operatorname{THH}(A) = \operatorname{THH}^{\mathbb{S}}(A)$.

There is also a bimodule formula

$$\operatorname{THH}^R(A) \simeq A \wedge_{A \wedge_R A^{\text{op}}}^{\mathbf{L}} A.$$

Here $A^e = A \wedge_R A^{\text{op}}$ is the enveloping algebra.

For the unit algebra:

$$\operatorname{THH}^R(R) \simeq R.$$

If A is commutative, multiplication on the cyclic bar construction makes $\operatorname{THH}^R(A)$ a commutative A -algebra.

23.2 The circle action

The cyclic operators give a canonical circle action on realization:

$$S^1 \curvearrowright \operatorname{THH}^R(A).$$

Warning 23.2 (The circle action is not yet cyclotomic structure). Topological cyclic homology also needs cyclotomic Frobenius maps:

$$\text{cyclic object} \implies S^1\text{-spectrum} \quad \text{from} \quad \text{cyclotomic spectrum} \implies \text{TC}.$$

23.3 Factorization homology and tensoring over spaces

For an E_1 -algebra A ,

$$\int_{S^1} A \simeq \operatorname{THH}(A).$$

For commutative A , tensoring over spaces is characterized by

$$\operatorname{Map}_{\operatorname{CAlg}_R}(X \otimes A, B) \simeq \operatorname{Map}_{\operatorname{Top}}(X, \operatorname{Map}_{\operatorname{CAlg}_R}(A, B)).$$

Then

$$\mathrm{THH}^R(A) \simeq S^1 \otimes A \quad (A \in \mathrm{CAlg}_R).$$

This formula uses commutativity.

Choosing a basepoint in S^1 gives maps

$$A \longrightarrow S^1 \otimes A \longrightarrow A$$

whose composite is the identity. The second map is induced by $S^1 \rightarrow *$.

The cyclic bar, derived Hochschild, and factorization-homology models agree under the stated hypotheses [Lur17].

24 TAQ and the Cotangent Complex

24.1 Tangent categories and linearization

For an ∞ -category $\mathcal{D}$, set

$$T_B \mathcal{D} = \mathbf{Sp}(\mathcal{D}_{/B})$$

and define the cotangent object by

$$L_B = \Sigma_{B,+}^\infty(\mathrm{id}_B) \in T_B \mathcal{D}.$$

These assemble to

$$L: \mathcal{D} \longrightarrow T\mathcal{D}, \quad B \longmapsto (B, L_B).$$

For $\mathcal{D} = \mathrm{CAlg}_R$, the tangent fiber at an algebra A is equivalent to the category of A -modules:

$$T_A \mathrm{CAlg}_R \simeq \mathrm{Mod}_A.$$

The map $R \rightarrow A$ determines the relative object $L_{A/R}$.

24.2 Cotangent specialization, square-zero extensions, and derivations

Fix a map of E_∞ rings $R \rightarrow A$ and an A -module M . There is a split square-zero extension

$$A \oplus M \longrightarrow A.$$

Underlying A -module: $A \oplus M$. Multiplication, schematically:

$$(a, m)(a', m') = (aa', am' + a'm), \quad mm' = 0.$$

A derivation of A over R with values in M is a section of the projection in commutative R -algebras:

$$\begin{array}{ccc} & & A \oplus M \\ & \nearrow d & \downarrow \\ A & \xlongequal{\quad} & A. \end{array}$$

Write $\mathrm{Der}_R(A, M)$ for the derived *space* of these sections. An A -module represents it.

Definition 24.1 (Cotangent complex and TAQ). The relative cotangent complex $L_{A/R} \in \text{Mod}_A$ is characterized by a natural equivalence

$$\text{Map}_{\text{Mod}_A}(L_{A/R}, M) \simeq \text{Der}_R(A, M).$$

Its underlying A -module is also written

$$\text{TAQ}^R(A) = L_{A/R}.$$

With coefficients, the associated homology and cohomology spectra may be written

$$\text{TAQ}^R(A; M) = M \wedge_A L_{A/R}, \quad \text{TAQ}_R(A; M) = \underline{\text{Map}}_A(L_{A/R}, M).$$

The mapping-spectrum refinement is

$$\text{Der}_R(A, M) \simeq \Omega^\infty \underline{\text{Map}}_A(L_{A/R}, M).$$

For a composable sequence $R \rightarrow A \rightarrow B$, there is a transitivity cofiber sequence of B -modules

$$B \wedge_A L_{A/R} \longrightarrow L_{B/R} \longrightarrow L_{B/A}.$$

For a free commutative algebra $A = \mathbb{P}_R(N)$, under the standard presentability hypotheses,

$$L_{A/R} \simeq A \wedge_R N.$$

24.3 Nonsplit extensions and Postnikov invariants

With the displayed shift convention, a derivation

$$\eta: L_{A/R} \longrightarrow \Sigma M.$$

gives a map $A \rightarrow A \oplus \Sigma M$ and the pullback

$$\begin{array}{ccc} A^\eta & \longrightarrow & A \\ \downarrow & & \downarrow \eta \\ A & \xrightarrow{0} & A \oplus \Sigma M. \end{array}$$

Its fiber over A is M .

For a connective commutative ring spectrum A , the Postnikov tower

$$\cdots \longrightarrow \tau_{\leq n} A \longrightarrow \tau_{\leq n-1} A \longrightarrow \cdots \longrightarrow \tau_{\leq 0} A$$

is built from square-zero extensions, one stage at a time. At the n th map, extend by the appropriately shifted $\pi_n A$ -module. A TAQ derivation represents the k -invariant. The extension need not split.

25 Algebraic K -Theory of Stable Categories

25.1 The Grothendieck group

Let $\mathcal{C}$ be an essentially small stable ∞ -category. Its Grothendieck group $K_0(\mathcal{C})$ is generated by symbols $[X]$ for objects $X \in \mathcal{C}$, subject to the relations

$$[Y] = [X] + [Z]$$

for every cofiber sequence

$$X \longrightarrow Y \longrightarrow Z.$$

In particular,

$$[0] = 0, \quad [X \oplus Y] = [X] + [Y], \quad [\Sigma X] = -[X].$$

For a ring spectrum R , usually use

$$\mathrm{Perf}(R) = (\mathrm{Mod}_R)^\omega,$$

the compact, or perfect, R -modules. Then

$$K_0(R) = K_0(\mathrm{Perf}(R)).$$

In a small triangulated category $\mathcal{T}$, a strictly full triangulated subcategory $\mathcal{A} \subseteq \mathcal{T}$ is *dense* if every object of $\mathcal{T}$ is a retract of an object of $\mathcal{A}$. Under the stated hypotheses, Thomason's classification gives a bijection

$$\left\{ \begin{array}{c} \text{strictly full dense triangulated} \\ \text{subcategories } \mathcal{A} \subseteq \mathcal{T} \end{array} \right\} \longleftrightarrow \{\text{subgroups of } K_0(\mathcal{T})\}.$$

The subgroup is the image of $K_0(\mathcal{A}) \rightarrow K_0(\mathcal{T})$.

26 Universal Invariants and Noncommutative Motives

26.1 Morita invariance, additivity, and localization

Let $\mathbf{Cat}_\infty^{\mathrm{perf}}$ be the ∞ -category of small idempotent-complete stable ∞ -categories and exact functors.

Definition 26.1 (Additive and localizing invariants). Let $E: \mathbf{Cat}_\infty^{\mathrm{perf}} \rightarrow \mathcal{D}$ take values in a stable presentable ∞ -category.

1. An *additive invariant* is Morita invariant, preserves filtered colimits, and takes a split-exact sequence

$$\mathcal{A} \longrightarrow \mathcal{B} \longrightarrow \mathcal{C}$$

to a direct-sum decomposition $E(\mathcal{B}) \simeq E(\mathcal{A}) \oplus E(\mathcal{C})$.

2. A *localizing invariant* satisfies the corresponding conditions for every exact, or Verdier, sequence, taking it to a cofiber sequence

$$E(\mathcal{A}) \longrightarrow E(\mathcal{B}) \longrightarrow E(\mathcal{C}).$$

Connective and nonconnective K -theory are respectively universal additive and localizing invariants, expressed by

$$\mathbf{Cat}_\infty^{\mathrm{perf}} \xrightarrow{U_{\mathrm{add}}} \mathrm{Mot}_{\mathrm{add}}, \quad \mathbf{Cat}_\infty^{\mathrm{perf}} \xrightarrow{U_{\mathrm{loc}}} \mathrm{Mot}_{\mathrm{loc}}.$$

Precomposition gives the corresponding universal properties.

26.2 Waldhausen categories and the $S_\bullet$ -construction

A Waldhausen category has cofibrations and weak equivalences compatible with pushout. Stable ∞ -categories carry the canonical structure in which all maps are ingressive.

Let $\text{Ar}[n]$ be the arrow category of the ordered set $[n] = \{0 < 1 < \dots < n\}$. An object of $S_n\mathcal{C}$ can be presented as a diagram

$$X: \text{Ar}[n] \longrightarrow \mathcal{C}$$

such that

$$X_{ii} \simeq 0$$

and every square

$$\begin{array}{ccc} X_{ij} & \longrightarrow & X_{ik} \\ \downarrow & & \downarrow \\ 0 \simeq X_{jj} & \longrightarrow & X_{jk} \end{array} \quad (i \leq j \leq k)$$

is a pushout. In a stable category it is also a pullback. The row

$$0 = X_{00} \longrightarrow X_{01} \longrightarrow \dots \longrightarrow X_{0n}$$

is a filtration with subquotients X_{ij} .

The connective K -theory space is

$$\Omega |(S_\bullet\mathcal{C})^\simeq|.$$

and iteration gives the spectrum $K(\mathcal{C})$, with

$$\pi_0 K(\mathcal{C}) \cong K_0(\mathcal{C}),$$

26.3 THH of stable categories and the cyclotomic trace

Topological Hochschild homology is Morita invariant and extends from ring spectra to small spectral or stable categories. If $\mathcal{C}$ is a spectral category, the degree- n cyclic object has the schematic form

$$\bigvee_{x_0, \dots, x_n} \mathcal{C}(x_0, x_1) \wedge \mathcal{C}(x_1, x_2) \wedge \dots \wedge \mathcal{C}(x_n, x_0).$$

Deriving and realizing gives $\text{THH}(\mathcal{C})$; for $\mathcal{C} = \text{Perf}(A)$ this is $\text{THH}(A)$.

The topological Dennis trace is a natural transformation

$$K(\mathcal{C}) \longrightarrow \text{THH}(\mathcal{C}).$$

The cyclotomic structure on THH gives the cyclotomic trace

$$K(\mathcal{C}) \longrightarrow \text{TC}(\mathcal{C}).$$

26.4 Corepresenting K -theory

Let $\mathbb{S}^\omega = \text{Perf}(\mathbb{S})$ be the small stable category of finite spectra. The tensor unit motive corepresents K -theory:

$$\text{Map}_{\text{Mot}_{\text{add}}}(U_{\text{add}}(\mathbb{S}^\omega), U_{\text{add}}(\mathcal{A})) \simeq K(\mathcal{A}),$$

and the analogous mapping spectrum in localizing motives recovers nonconnective K -theory.

For smooth and proper $\mathcal{A}$,

$$\text{Map}_{\text{Mot}}(U(\mathcal{A}), U(\mathcal{B})) \simeq K(\mathcal{A}^{\text{op}} \otimes \mathcal{B}).$$

27 Picard Spaces and Generalized Thom Spectra

27.1 Invertible objects

Let $\mathcal{C}$ be a symmetric monoidal ∞ -category. An object $L \in \mathcal{C}$ is tensor-invertible if there is an object L^{-1} and an equivalence

$$L \otimes L^{-1} \simeq \mathbf{1}.$$

The full maximal subgroupoid of invertible objects and their equivalences is the Picard space

$$\mathrm{Pic}(\mathcal{C}) = (\mathcal{C}^\simeq)^\times.$$

It is a grouplike E_∞ -space, so it is the infinite loop space of a connective Picard spectrum $\mathrm{pic}(\mathcal{C})$.

For an E_∞ ring R , write

$$\mathrm{Pic}(R) = \mathrm{Pic}(\mathrm{Mod}_R).$$

Its loop space at the unit is

$$\Omega_R \mathrm{Pic}(R) \simeq \mathrm{Aut}_R(R) \simeq GL_1(R).$$

Its homotopy groups satisfy

$$\begin{aligned} \pi_0 \mathrm{pic}(R) &= \mathrm{Pic}(R), \\ \pi_1 \mathrm{pic}(R) &= (\pi_0 R)^\times, \\ \pi_i \mathrm{pic}(R) &= \pi_{i-1} R, \quad i \geq 2. \end{aligned}$$

27.2 Descent

The functor $R \mapsto \mathrm{Pic}(R)$ satisfies derived étale descent. For connective R , its spectral sequence starts with étale cohomology of the homotopy sheaves of $\mathrm{pic}(R)$.

27.3 Picard-valued Thom twists

The connected component of the unit is exactly

$$\mathrm{Pic}(R)_{[R]} \simeq R\text{-line} \simeq BGL_1(R).$$

A general Picard-valued twist is a map

$$f: X \longrightarrow \mathrm{Pic}(R)$$

with Thom module

$$M_R f = \mathrm{colim}(X \xrightarrow{f} \mathrm{Pic}(R) \hookrightarrow \mathrm{Mod}_R).$$

27.4 The complex J -homomorphism and MUP

Complex vector bundles give a refinement of the stable J -homomorphism

$$J_{\mathbb{C}}: BU \times \mathbb{Z} \longrightarrow \mathrm{Pic}(\mathbb{S}).$$

The integer component records virtual complex rank, hence even suspension degree. The Thom spectrum is periodic complex bordism MUP . For the underlying spectrum:

$$MUP \simeq \bigvee_{n \in \mathbb{Z}} \Sigma^{2n} MU.$$

This holds for spectra (or MU -modules). It is not a ring-spectrum splitting. Multiplication adds the wedge indices.

28 Brauer Spaces and Morita-Invertible Categories

28.1 Morita-invertible module categories

An associative R -algebra A is *Azumaya* if Mod_A is invertible as an R -linear category. Under the usual hypotheses, A is dualizable and faithful and the map

$$A \wedge_R A^{\text{op}} \longrightarrow \underline{\text{End}}_R(A)$$

is an equivalence.

The Brauer space $\text{Br}(R)$ is the maximal ∞ -groupoid of invertible R -linear categories; its components are Morita classes of Azumaya algebras.

28.2 Étale descent and low homotopy groups

The Brauer space satisfies derived étale descent. For a connective E_∞ ring R , its low homotopy groups are

$$\begin{aligned} \pi_0 \text{Br}(R) &\cong H_{\text{ét}}^1(\text{Spec } \pi_0 R; \mathbb{Z}) \oplus H_{\text{ét}}^2(\text{Spec } \pi_0 R; \mathbb{G}_m), \\ \pi_1 \text{Br}(R) &\cong H_{\text{ét}}^0(\text{Spec } \pi_0 R; \mathbb{Z}) \oplus H_{\text{ét}}^1(\text{Spec } \pi_0 R; \mathbb{G}_m), \\ \pi_2 \text{Br}(R) &\cong (\pi_0 R)^\times, \\ \pi_k \text{Br}(R) &\cong \pi_{k-2} R, \end{aligned} \quad k \geq 3.$$

Here $\pi_1 \text{Br}(R) \cong \pi_0 \text{Pic}(R)$, and the H^2 term is cohomological Brauer theory.

Understanding / Intuition

Units: invertible endomorphisms of the monoidal unit. Picard objects: invertible modules. Brauer objects: invertible module categories.

28.3 Looping and delooping relations

At the basepoint Mod_R ,

$$\Omega_{\text{Mod}_R} \text{Br}(R) \simeq \text{Pic}(R).$$

Looping once more gives

$$\Omega_R \text{Pic}(R) \simeq GL_1(R).$$

For the corresponding components:

$$\text{Pic}(R)_{[R]} \simeq BGL_1(R), \quad \text{Br}(R)_{[\text{Mod}_R]} \simeq B\text{Pic}(R).$$

These equivalences are componentwise.

29 Bipermutative Categories and ku

Let $\mathcal{V}_{\mathbb{C}}$ be the topological groupoid of finite-dimensional complex inner-product spaces and unitary isomorphisms. It has direct sum and tensor product

$$(\mathcal{V}_{\mathbb{C}}, \oplus, 0), \quad (\mathcal{V}_{\mathbb{C}}, \otimes, \mathbb{C}),$$

with coherent distributivity equivalences

$$V \otimes (W \oplus U) \cong (V \otimes W) \oplus (V \otimes U),$$

and similarly in the other variable. After coherent strictification, this is a bipermutative, or bisymmetric monoidal, category.

The additive machine starts from

$$\coprod_{n \geq 0} BU(n),$$

whose group completion is

$$\mathbb{Z} \times BU \simeq \Omega^{\infty} ku.$$

Tensor product makes the resulting connective spectrum ku an E_{∞} ring spectrum.

The homotopy ring is

$$\pi_* ku \cong \mathbb{Z}[u], \quad |u| = 2,$$

where u is the Bott class. Inverting u gives periodic complex K -theory:

$$KU \simeq ku[u^{-1}].$$

PART V

Day 5: Localization and Spectral Algebraic Geometry

Convention: tensor products, mapping spaces, limits, and colimits are derived unless stated otherwise.

30 Bousfield Localization

30.1 Equivalences detected by a homology theory

Let E be a spectrum. A map $f: X \rightarrow Y$ is an E -equivalence if

$$E \wedge f: E \wedge X \longrightarrow E \wedge Y$$

is an equivalence. By stability, equivalently,

$$E \wedge \operatorname{cof}(f) \simeq 0.$$

For example, $H\mathbb{Z}/2$ -equivalences induce isomorphisms on mod-2 homology.

Arithmetic fracture square: separate the rational and primary parts of a spectrum.
Adams resolution: approximate it by objects built from repeated copies of E .

30.2 Local objects and local equivalences

Let $\mathcal{C}$ be an ∞ -category with mapping spaces, and let S be a collection of maps in $\mathcal{C}$.

Definition 30.1 (Local object). An object $Z \in \mathcal{C}$ is S -local if, for every map $s: A \rightarrow B$ in S , precomposition induces an equivalence

$$s^*: \operatorname{Map}_{\mathcal{C}}(B, Z) \xrightarrow{\simeq} \operatorname{Map}_{\mathcal{C}}(A, Z).$$

Definition 30.2 (Local equivalence). A map $g: X \rightarrow Y$ is an S -local equivalence if every S -local object Z sees it as an equivalence:

$$g^*: \operatorname{Map}_{\mathcal{C}}(Y, Z) \xrightarrow{\simeq} \operatorname{Map}_{\mathcal{C}}(X, Z).$$

Definition 30.3 (Localization functor). An S -localization is a functor $L: \mathcal{C} \rightarrow \mathcal{C}$ with a natural map $\eta_X: X \rightarrow LX$ such that LX is S -local and η_X is an S -local equivalence for every X .

The universal property gives

$$\operatorname{Map}_{\mathcal{C}}(LX, Z) \simeq \operatorname{Map}_{\mathcal{C}}(X, Z) \quad (Z \text{ local}).$$

It also gives $L^2 \simeq L$. The full subcategory $\mathcal{C}^{\operatorname{loc}} \subseteq \mathcal{C}$ is reflective:

$$\mathcal{C} \begin{array}{c} \xrightarrow{L} \\ \xleftarrow{i} \end{array} \mathcal{C}^{\operatorname{loc}} \quad L \dashv i,$$

where i is fully faithful. For presentable $\mathcal{C}$ and a set S , the local subcategory is presentable.

30.3 Basic examples

Example: $p: \mathbb{Z} \rightarrow \mathbb{Z}$ in abelian groups. An abelian group A is local for this map iff multiplication by p is an automorphism. These are $\mathbb{Z}[1/p]$ -modules. Invert every nonzero integer to get rational vector spaces. Inverting just one prime does not give rationalization.

For commutative monoids, the map

$$\mathbb{N} \longrightarrow \mathbb{Z}$$

gives group completion. Local objects: commutative monoids in which every element is invertible. Same idea for E_∞ -spaces: group completion takes a homotopy-commutative monoid to a grouplike one.

For spaces, localization at $S^{n+1} \rightarrow *$ gives n -truncated spaces. Indeed, for an n -truncated space Z , evaluation gives

$$\mathrm{Map}(*, Z) \xrightarrow{\simeq} \mathrm{Map}(S^{n+1}, Z).$$

Localization at $\emptyset \rightarrow *$ has only contractible local objects. Localizations defined by maps involving $K(\mathbb{Z}/p, n)$ isolate primary information. For nilpotent spaces, Sullivan localization can be described by inverting selected degree maps on spheres; rationalization inverts all primes, while p -localization inverts the primes different from p .

Theorem 30.4 (Sullivan rational homotopy theory). *After imposing the standard nilpotence and finite rational-type hypotheses, the rational homotopy category of simply connected spaces is contravariantly equivalent to the homotopy category of 1-connected Sullivan commutative differential graded $\mathbb{Q}$ -algebras of finite type. Here 1-connected means $H^0 \cong \mathbb{Q}$ and $H^1 = 0$. The functor from spaces is the polynomial de Rham algebra A_{PL} .*

For nilpotent finite-type X ,

$$\pi_n(X_{\mathbb{Q}}) \cong \pi_n(X) \otimes \mathbb{Q} \quad (n \geq 1),$$

with the usual fundamental-group convention.

In spectra, rationalization is smashing:

$$L_{H\mathbb{Q}}X \simeq H\mathbb{Q} \wedge X.$$

Completion at p is a different operation; see below.

30.4 The small-object construction

Replace $S = \{A_\alpha \rightarrow B_\alpha\}$ by cofibrations between cofibrant objects. Given

$$\begin{array}{ccc} A_\alpha & \longrightarrow & X \\ \downarrow & & \downarrow \\ B_\alpha & \longrightarrow & X \cup_{A_\alpha} B_\alpha, \end{array}$$

attaching B_α extends the map $A_\alpha \rightarrow X$.

For $n \geq 0$, also use the pushout-product

$$(A_\alpha \otimes \Delta^n) \cup_{A_\alpha \otimes \partial \Delta^n} (B_\alpha \otimes \partial \Delta^n) \longrightarrow B_\alpha \otimes \Delta^n.$$

Use $n = 1, 2, \dots$ for the higher homotopies. For a local object Z ,

$$\mathrm{Map}(B_\alpha \cup_{A_\alpha}^h B_\alpha, Z) \simeq \mathrm{Map}(B_\alpha, Z) \times_{\mathrm{Map}(A_\alpha, Z)}^h \mathrm{Map}(B_\alpha, Z).$$

Apply the small-object argument to these maps together with the original generating acyclic cofibrations. This gives a natural map

$$X \longrightarrow L_S X.$$

Under the usual smallness hypotheses, the target is fibrant and local.

30.5 Left Bousfield localization of a model category

Let $\mathcal{M}$ be a model category. A *left Bousfield localization* $L_S \mathcal{M}$, when it exists, has

$$\begin{aligned} \text{cofibrations in } L_S \mathcal{M} &= \text{cofibrations in } \mathcal{M}, \\ \text{weak equivalences in } L_S \mathcal{M} &= S\text{-local equivalences,} \end{aligned}$$

and its fibrations are determined by the lifting axioms. Its fibrant objects are the fibrant S -local objects of $\mathcal{M}$. The identity functor gives a Quillen adjunction

$$\mathcal{M} \begin{array}{c} \xrightarrow{\text{id}} \\ \xleftarrow{\text{id}} \end{array} L_S \mathcal{M}$$

with the left adjoint pointing toward the localization.

31 Existence and Monoidal Localization

31.1 Combinatorial model categories

Definition 31.1 (Combinatorial model category). A model category is *combinatorial* if its underlying category is locally presentable and its model structure is cofibrantly generated.

The following recognition theorem is often called Smith's theorem.

Theorem 31.2 (Smith recognition, schematic form). *Let $\mathcal{C}$ be locally presentable, let I be a set of maps, and let W be an accessible, accessibly embedded class of maps. Assume:*

1. *W has two-out-of-three and is closed under retracts;*
2. *$I\text{-inj} \subseteq W$;*
3. *$W \cap \text{cof}(I)$ is closed under pushouts and transfinite composition.*

Then $\mathcal{C}$ admits a combinatorial model structure whose weak equivalences are W and whose cofibrations are $\text{cof}(I)$.

For a left proper combinatorial simplicial model category these conditions produce $L_S \mathcal{M}$.

For a presentable ∞ -category, can localize at a set of maps. The new equivalences form the strongly saturated class generated by the set. Closure operations: equivalences, two-out-of-three, retracts, pushouts, and colimits in the arrow category.

31.2 Stable localizations

Let $\mathcal{C}$ be a stable ∞ -category and let $L: \mathcal{C} \rightarrow \mathcal{C}$ be a localization. The following conditions are equivalent:

1. the full subcategory of local objects is stable;
2. L is exact;

3. the L -acyclic objects, those X with $LX \simeq 0$, form a stable subcategory;
4. the local equivalences are closed under suspension and desuspension.

Take the fiber of the localization map. This gives the acyclization functor Γ and a natural fiber sequence:

$$\Gamma X \longrightarrow X \longrightarrow LX.$$

The kernel is the acyclic subcategory.

Localization at a spectrum E is stable because $E \wedge -$ is exact. More generally, a localization generated by maps closed under suspension is stable. The coconnective truncation $\tau_{\leq n}$ on spectra is a localization, but not a stable one: its local objects are not closed under both suspension and desuspension.

31.3 Localization and completion at a prime

For spectra, keep these three operations separate:

- p -localization inverts every prime other than p : $X_{(p)} \simeq X \wedge \mathbb{S}_{(p)}$.
- Inverting p gives $X[1/p]$.
- p -completion keeps the information detected by mod- p homology.

For bounded-below X , a model for derived p -completion is

$$X_p^\wedge \simeq \operatorname{holim}_r X/p^r, \quad X/p^r = \operatorname{cof}(p^r: X \rightarrow X).$$

With the same boundedness assumptions, this is localization at $H\mathbb{F}_p$. Without them, the two can differ.

p -completion is not generally smashing.

31.4 Smashing localizations

Let $\mathcal{C}$ be a presentably symmetric monoidal stable ∞ -category, and assume tensoring preserves colimits separately in each variable.

Definition 31.3 (Smashing localization). A localization L is *smashing* if there is a natural equivalence

$$LX \simeq X \otimes L\mathbf{1}$$

for every $X \in \mathcal{C}$.

Here $L\mathbf{1}$ is an idempotent commutative algebra object:

$$L\mathbf{1} \otimes L\mathbf{1} \simeq L\mathbf{1}.$$

Conversely, tensoring with an idempotent algebra gives a smashing localization.

Examples: rationalization and finite localizations. Postnikov truncation is not smashing. Neither is p -completion in general.

31.5 The tensor product after localization

Assume local equivalences are preserved by tensoring with any object. If

$$L: \mathcal{C} \rightleftarrows \mathcal{C}^{\operatorname{loc}} : i,$$

the tensor product in the local category is

$$X \otimes_{\operatorname{loc}} Y := L(iX \otimes iY), \quad \mathbf{1}_{\operatorname{loc}} = L\mathbf{1}.$$

Then L is symmetric monoidal and i is lax symmetric monoidal.

32 Local Duality and Completion

32.1 Torsion, local, and complete modules

Let R be an E_∞ -ring, and let $I \subseteq \pi_0 R$ be a finitely generated ideal. Write R/I for a chosen derived quotient or Koszul object. Three subcategories of Mod_R :

Definition 32.1 (Torsion, local, and complete). An R -module M is

1. *I -torsion* if it belongs to the localizing tensor ideal generated by R/I ;
2. *I -local* if $\mathrm{Map}_R(T, M) \simeq *$ for every I -torsion module T ;
3. *I -complete* if $\mathrm{Map}_R(N, M) \simeq *$ for every I -local module N .

Let Γ_I be the right adjoint to the inclusion of the torsion subcategory, and let L_I be localization away from torsion. There is a natural fiber sequence

$$\Gamma_I M \longrightarrow M \longrightarrow L_I M. \quad (5.1)$$

The completion functor Λ^I is left adjoint to the inclusion of complete modules, and local duality gives

$$\mathrm{Map}_R(\Gamma_I M, N) \simeq \mathrm{Map}_R(M, \Lambda^I N) \quad (5.2)$$

under the usual compact-generation hypotheses.

For an ordinary Noetherian ring and a finitely generated ideal, derived completion is represented by

$$\Lambda^I M \simeq \mathrm{holim}_r (M \otimes_R R/I^r). \quad (5.3)$$

Under the usual finiteness hypotheses, the analogous spectral formula is localization at R/I .

If $I = (x)$ is principal, the local term is often

$$L_I M \simeq M \otimes_R R[x^{-1}].$$

For a general ideal, use the derived Čech object or the open complement of $V(I)$. A single ring $R[I^{-1}]$ is not enough.

32.2 Adams towers and cobar resolutions

Let A be an E_1 -algebra object in a stable symmetric monoidal category, with unit $\mathbf{1} \rightarrow A$, and set

$$J = \mathrm{fib}(\mathbf{1} \longrightarrow A).$$

Tensor powers of J give the Adams tower

$$\cdots \longrightarrow J^{\otimes 3} \otimes X \longrightarrow J^{\otimes 2} \otimes X \longrightarrow J \otimes X \longrightarrow X.$$

The successive A -nilpotent approximations are

$$C_n X = \mathrm{cof}(J^{\otimes(n+1)} \otimes X \longrightarrow X).$$

Take the inverse limit for completion:

$$X_A^\wedge := \lim_n C_n X.$$

The same data give the augmented cobar object

$$X \longrightarrow A \otimes X \rightrightarrows A^{\otimes 2} \otimes X \rightrightarrows A^{\otimes 3} \otimes X \longrightarrow \dots$$

Under the standard comparison hypotheses, its totalization gives the same A -nilpotent completion:

$$X_A^\wedge \simeq \mathrm{Tot}(A^{\otimes(\bullet+1)} \otimes X), \quad X \longrightarrow X_A^\wedge. \quad (5.4)$$

This is the Adams spectral sequence. For descendable A , the completion map is an equivalence.

For $A = H\mathbb{F}_p$ and bounded-below spectra, this is derived p -completion. This is what occurs in the classical mod- p Adams spectral sequence.

32.3 Geometry of local duality

Let $X = \mathrm{Spec} R$, let $Z = V(I)$, and let $U = X \setminus Z$. Geometric versions of the three module categories:

algebra	geometry
I -torsion	quasi-coherent sheaves supported on Z
I -local	quasi-coherent sheaves on U
I -complete	quasi-coherent sheaves on $\widehat{X}_Z$.

Here $\widehat{X}_Z$ is the formal completion of X along Z .

Under the standard descent and finiteness assumptions, formal gluing is

$$\begin{array}{ccc} \mathrm{QCoh}(X) & \longrightarrow & \mathrm{QCoh}(U) \\ \downarrow & & \downarrow \\ \mathrm{QCoh}(\widehat{X}_Z) & \longrightarrow & \mathrm{QCoh}(\widehat{X}_Z^\circ), \end{array} \quad (5.5)$$

where $\widehat{X}_Z^\circ$ is the punctured formal neighborhood.

33 Chromatic Localization

Fix a prime p and work in p -local spectra.

33.1 Morava K -theories and height

For $0 < n < \infty$, Morava K -theory has coefficients

$$K(n)_* \cong \mathbb{F}_p[v_n^{\pm 1}], \quad |v_n| = 2(p^n - 1). \quad (5.6)$$

The endpoint conventions are

$$K(0) = H\mathbb{Q}, \quad K(\infty) = H\mathbb{F}_p.$$

Every $K(n)$ -module splits as a wedge of suspensions of $K(n)$.

For a nonzero finite p -local spectrum F , define

$$\mathrm{type}(F) = n$$

when $K(m)_*F = 0$ for $m < n$ and $K(n)_*F \neq 0$.

33.2 Nilpotence and thick subcategories

Let $f: F \rightarrow X$, where F is finite and X is p -local. If $K(n)_*f = 0$ for every $0 \leq n \leq \infty$, then for some N ,

$$f^{\wedge N} \simeq 0.$$

For a ring spectrum, MU_* detects nilpotent elements.

Let

$$\mathcal{C}_n = \{F \in \mathbf{Sp}_{(p)}^{\text{fin}} : K(n-1)_*F = 0\} \quad (n \geq 1),$$

and put $\mathcal{C}_0 = \mathbf{Sp}_{(p)}^{\text{fin}}$. Mitchell's existence theorem gives finite p -local spectra of every type. Hence

$$\mathcal{C}_0 \supsetneq \mathcal{C}_1 \supsetneq \mathcal{C}_2 \supsetneq \cdots. \quad (5.7)$$

Theorem 33.1 (Hopkins–Smith). *Every nonzero thick subcategory of finite p -local spectra is exactly one of the subcategories $\mathcal{C}_n$.*

33.3 Periodic self-maps and telescopes

Let F have type n . A v_n -self-map is a map

$$v: \Sigma^d F \longrightarrow F$$

that acts invertibly on $K(n)_*F$ and nilpotently at the other relevant heights. The periodicity theorem gives such a self-map. Sufficiently high iterates satisfy the theorem's asymptotic uniqueness and naturality statements.

Desuspend and iterate v to form the telescope

$$T(n) = \text{colim}(F \xrightarrow{v} \Sigma^{-d} F \xrightarrow{v} \Sigma^{-2d} F \longrightarrow \cdots). \quad (5.8)$$

Its Bousfield class does not depend on the finite type- n spectrum or the choice of periodic self-map.

The telescope conjecture compared two localizations:

$$L_{T(n)} \stackrel{?}{\simeq} L_{K(n)}. \quad (5.9)$$

It holds for $n \leq 1$ and fails for $n \geq 2$.

33.4 The chromatic tower

Let

$$L_n = L_{K(0) \vee K(1) \vee \cdots \vee K(n)}.$$

This is the usual $E(n)$ -localization. It is smashing. The maps of Bousfield classes give the chromatic tower:

$$\cdots \longrightarrow L_n X \longrightarrow L_{n-1} X \longrightarrow \cdots \longrightarrow L_0 X.$$

Its n th monochromatic layer is

$$M_n X = \text{fib}(L_n X \longrightarrow L_{n-1} X).$$

Theorem 33.2 (Chromatic convergence). *For a finite p -local spectrum F , the natural map*

$$F \longrightarrow \text{holim}_n L_n F$$

is an equivalence.

33.5 The $K(n)$ -local sphere

Choose a cofinal tower of generalized Moore spectra M_I with

$$BP_*M_I \cong BP_*/I, \quad I = (p^{i_0}, v_1^{i_1}, \dots, v_{n-1}^{i_{n-1}}),$$

where the exponents range through a cofinal system. Under the standard finite-type hypotheses, one has

$$L_{K(n)}X \simeq \operatorname{holim}_I (L_nX \wedge M_I). \quad (5.10)$$

See [HS99].

Let E_n be Morava E -theory. Its coefficient ring is

$$(E_n)_0 \cong W(\mathbb{F}_{p^n})[[u_1, \dots, u_{n-1}]], \quad (E_n)_* \cong (E_n)_0[u^{\pm 1}], \quad |u| = -2.$$

The extended Morava stabilizer group $\mathbb{G}_n$ acts on E_n , and the Devinatz–Hopkins theorem identifies

$$L_{K(n)}\mathbb{S} \simeq E_n^{h\mathbb{G}_n}. \quad (5.11)$$

For a finite spectrum X , the associated continuous homotopy-fixed-point spectral sequence is

$$H_c^s(\mathbb{G}_n; \pi_t(E_n \wedge X)) \implies \pi_{t-s}L_{K(n)}X. \quad (5.12)$$

At height 1, this gives a description using p -adic K -theory and the action of $\mathbb{Z}_p^\times$.

34 Spectral Algebraic Geometry

34.1 Flat maps and fpqc descent

Let $f: A \rightarrow B$ be a map of E_∞ -rings.

Definition 34.1 (Flat map). The map f is *flat* if $\pi_0 B$ is flat over $\pi_0 A$ and the natural maps

$$\pi_n A \otimes_{\pi_0 A} \pi_0 B \longrightarrow \pi_n B$$

are isomorphisms for every n . It is *faithfully flat* if, in addition, $\pi_0 A \rightarrow \pi_0 B$ is faithfully flat.

A faithfully flat map has an Amitsur cosimplicial algebra

$$B \rightrightarrows B \otimes_A B \rightrightarrows B \otimes_A B \otimes_A B \longrightarrow \dots$$

Faithfully flat descent gives

$$A \simeq \operatorname{Tot}(B^{\otimes_A(\bullet+1)}) \quad (5.13)$$

and, more strongly,

$$\operatorname{Mod}_A \simeq \operatorname{Tot}([q] \mapsto \operatorname{Mod}_{B^{\otimes_A(q+1)}}). \quad (5.14)$$

The fpqc topology is generated by faithfully flat quasi-compact covers.

34.2 Spectral affine schemes and stacks

Let

$$\mathrm{Aff}^{\mathrm{sp}} = \mathrm{CAlg}(\mathbf{Sp})^{\mathrm{op}}.$$

For an E_∞ -ring A , the spectral affine scheme $\mathrm{Spec} A$ is the functor

$$\mathrm{Spec} A(B) = \mathrm{Map}_{\mathrm{CAlg}(\mathbf{Sp})}(A, B). \quad (5.15)$$

Definition 34.2 (Spectral stack). A *spectral stack* for the fpqc topology is a functor

$$\mathcal{X}: \mathrm{CAlg}(\mathbf{Sp}) \longrightarrow \mathcal{S}$$

that satisfies fpqc descent. Other spectral Grothendieck topologies give corresponding notions of spectral stack.

For an affine spectral scheme,

$$\mathrm{QCoh}(\mathrm{Spec} A) \simeq \mathrm{Mod}_A. \quad (5.16)$$

Global functions give back the ring:

$$\Gamma(\mathrm{Spec} A, \mathcal{O}) \simeq A.$$

For a stack $\mathcal{X}$ presented by spectral affines $\mathrm{Spec} A_i \rightarrow \mathcal{X}$, get the quasi-coherent category by descent:

$$\mathrm{QCoh}(\mathcal{X}) \simeq \lim_{\mathrm{Spec} A \rightarrow \mathcal{X}} \mathrm{Mod}_A. \quad (5.17)$$

The structure sheaf has an E_∞ -ring of global sections $\Gamma(\mathcal{X}, \mathcal{O}_{\mathcal{X}})$.

For connective A , the classical truncation of $\mathrm{Spec} A$ is

$$t_0(\mathrm{Spec} A) = \mathrm{Spec} \pi_0 A.$$

The groups $\pi_i A$ give quasi-coherent sheaves on the truncation [BGH22].

34.3 Spherical monoid algebras

For a commutative monoid M , define its spherical monoid algebra by

$$\mathbb{S}[M] = \Sigma_+^\infty M.$$

This gives the spectral analogues

$$\mathbb{A}_{\mathbb{S}}^1 = \mathrm{Spec} \mathbb{S}[\mathbb{N}], \quad \mathbb{G}_{m, \mathbb{S}} = \mathrm{Spec} \mathbb{S}[\mathbb{Z}]. \quad (5.18)$$

The map $\mathbb{N} \rightarrow \mathbb{Z}$ induces the open inclusion of the spectral multiplicative group into the spectral affine line. Base change along $\mathbb{S} \rightarrow H\mathbb{Z}$ gives $H\mathbb{Z}[t]$ and $H\mathbb{Z}[t^{\pm 1}]$; their degree-zero homotopy rings are $\mathbb{Z}[t]$ and $\mathbb{Z}[t^{\pm 1}]$.

34.4 Formal spectra and formal groups

For an ideal $I \subseteq \pi_0 A$, write the formal spectrum as the functor given by the infinitesimal neighborhoods:

$$\mathrm{Spf}(A, I) \simeq \operatorname{colim}_r \mathrm{Spec}(A/I^r). \quad (5.19)$$

Its quasi-coherent sheaves are derived I -complete modules under the usual finiteness hypotheses.

The moduli stack of one-dimensional formal groups is $\mathcal{M}_{FG}$. A change of coordinate has the form

$$x' = a_1 x + a_2 x^2 + \cdots, \quad a_1 \in (\pi_0 A)^\times.$$

It is strict only when $a_1 = 1$. On an E_∞ -ring A , its functor of points is

$$\mathcal{M}_{FG}(A) \simeq \mathrm{FGroup}(A)^\simeq,$$

the maximal ∞ -groupoid of formal groups over A .

34.5 Preorientations and orientations

Let G be a one-dimensional formal group over an E_∞ -ring R , not necessarily connective. A preorientation is a based map

$$e: S^2 \longrightarrow G(\tau_{\geq 0} R).$$

The formal group has a dualizing line ω_G . The preorientation determines a Bott map

$$\beta_e: \omega_G \longrightarrow \Sigma^{-2} R. \quad (5.20)$$

Definition 34.3 (Orientation). The preorientation e is an *orientation* if the Bott map β_e is an equivalence.

An orientation identifies the cotangent and degree-2 periodicity lines [BGH22].

34.6 Periodic complex bordism and descent

Let MUP denote periodic complex bordism. The universal orientation gives an fpqc orientation chart

$$\mathrm{Spec} MUP \longrightarrow \mathcal{M}_{FG}^{\mathrm{or}},$$

where $\mathcal{M}_{FG}^{\mathrm{or}}$ is the spectral moduli stack of oriented formal groups. Its Čech nerve has terms

$$\mathrm{Spec}(MUP^{\otimes(q+1)}).$$

So compute global functions by the Amitsur totalization:

$$\Gamma(\mathcal{M}_{FG}^{\mathrm{or}}, \mathcal{O}) \simeq \mathrm{Tot}(MUP^{\otimes(\bullet+1)}) \simeq \mathbb{S}. \quad (5.21)$$

The descent spectral sequence is the periodic Adams–Novikov spectral sequence.

34.7 Elliptic curves and topological modular forms

An elliptic curve has a formal group at its identity section. This gives a map

$$\mathcal{M}_{Ell} \longrightarrow \mathcal{M}_{FG}.$$

Adding an orientation gives the pullback square

$$\begin{array}{ccc} \mathcal{M}_{Ell}^{\text{or}} & \longrightarrow & \mathcal{M}_{FG}^{\text{or}} \\ \downarrow & & \downarrow \\ \mathcal{M}_{Ell} & \longrightarrow & \mathcal{M}_{FG}. \end{array} \quad (5.22)$$

The sheaf of E_∞ -rings on the elliptic moduli stack is even and weakly 2-periodic on the smooth locus. For an even periodic affine chart $\text{Spec } R$,

$$\pi_{2k+1}R = 0, \quad \pi_{2k}R \cong (\pi_2R)^{\otimes_{\pi_0R} k},$$

where negative powers use the inverse line. Viewing $\mathcal{M}_{Ell}^{\text{or}}$ as the spectral enhancement carrying $\mathcal{O}^{\text{top}}$, its global sections give periodic topological modular forms:

$$\Gamma(\mathcal{M}_{Ell}^{\text{or}}, \mathcal{O}) \simeq \Gamma(\mathcal{M}_{Ell}, \mathcal{O}^{\text{top}}) \simeq TMF. \quad (5.23)$$

Notation: tmf , Tmf , and TMF for the connective, compactified, and periodic versions. Need the exact convention when discussing the boundary or connectivity.

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